

Tap Against Me Bible

Effects

Tapping Effects

Tap Against Me — Tap Effects System

Developer Handoff

Tap effects are cosmetic effects triggered at the location of each valid tap. They should work in standard local play and Infinity Taps without changing jump strength, gravity, movement speed, fruit spawns, coin rewards, or competitive outcomes.

Each player may equip one primary tap effect at a time. Some effects may also support optional color variants or intensity settings.

1. Pulsing Rings

A circular ring expands outward from the tap location and fades away.

Base Version

- Default color: Purple
- Price: **100 Hero Coins**
- One ring appears per tap.
- Ring expands smoothly before disappearing.
- Effect should remain lightweight enough for rapid tapping.

Custom Color Variants

- Price: **50 Hero Coins per unlocked color slot or color variant**
- Opens a color picker.
- Player can configure:
 - Hue
 - Saturation
 - Brightness
 - RGB values
 - Darkness
 - Opacity, within safe visibility limits

The custom color should be stored as part of the player's equipped cosmetic settings rather than creating a separate inventory item for every selected color.

RGB Pulse Variant

- Price: ****300 Hero Coins****
- Every tap selects a new color.
- Colors may:
 - Change randomly
 - Move progressively through the color spectrum
 - Alternate between a saved palette

Include a setting:

```
```text
Random RGB | Smooth Rainbow | Custom Palette
```
```

For accessibility, rapidly flashing colors should be avoided. The effect should change color per pulse without creating full-screen flashes.

2. Rain Tap Effect

Rain falls from above the tap location.

- Price: ****200 Hero Coins****
- Each tap creates a short localized rain shower.
- Rain droplets fall toward the nearest ground or terrain surface.
- Multiple taps can produce overlapping rain showers.
- Water should fade or evaporate after a short period rather than permanently modifying the world.

Rare Lightning Event

Each rain tap has a ****0.001% chance**** to generate a lightning strike.

That is approximately one lightning event per 100,000 taps on average, although random results will vary.

When triggered:

1. Clouds briefly form above the tap location.
2. Lightning strikes downward.
3. A short flash and thunder sound play.
4. The ground receives a temporary scorched mark.
5. Small smoke particles rise.

6. Nearby characters may perform a surprised animation.

The burn mark should disappear after a configured period and must not damage the player.

The effect should be purely visual, including in multiplayer.

3. Snow Tap Effect

Snow falls around the tap location.

- Price: ****200 Hero Coins****
- Each tap produces a small burst of snowflakes.
- Snow accumulates temporarily on valid ground surfaces.
- Repeated tapping in the same area increases the size of the snow pile.
- Snow slowly melts after the player stops adding to it.

Snow Accumulation

Track temporary snow mass by ground region or collision section.

Example states:

```
``text  
Light Dusting  
Small Pile  
Large Pile  
Maximum Pile  
``
```

The snow pile should have a maximum size so it cannot grow indefinitely or block gameplay.

Evil Snowman Event

When a snow pile reaches the required size, it gains a chance to produce an ****Evil Snowman****.

The snowman:

- Rises from the accumulated snow.
- Performs a short sinister or comedic animation.
- May look toward the player, laugh, throw harmless snow, or wander briefly.
- Eventually melts or collapses.
- Does not attack, damage, or interrupt the player.

The spawn chance should be checked when the pile crosses a size threshold rather than on every rendered snow particle.

4. Magma Tap Effect

Drops of magma or lava fall from the tap location.

- Price: ****200 Hero Coins****
- Each tap creates a small magma fall or molten splash.
- Lava briefly accumulates on the ground.
- Repeated tapping increases the size of the molten pool.
- The pool gradually flows outward, cools, and disappears.

Lava Accumulation

The pool may transition through:

```
``text
Small Splash
Molten Puddle
Large Pool
Cooling Pool
Obsidian Remains
``
```

The lava must remain cosmetic. It cannot damage players, modify movement, or block paths.

Obsidian Golem Event

When a large amount of lava accumulates, there is a chance for an ****Obsidian Golem**** to rise from it.

The golem:

- Emerges from the molten pool.
- Cools from orange magma to dark obsidian.
- Performs a short roar, pose, or ground-pounding animation.
- May follow the player briefly or remain as a temporary background character.
- Crumbles into harmless obsidian fragments before disappearing.

The golem does not provide coins, gameplay power, or competitive benefits.

5. Fire Tap Effect

A small fire burst appears at the tap location.

- Price: **75 Hero Coins**
- Designed as a lower-cost first cosmetic.
- Each tap produces:
 - A brief flame
 - Sparks or embers
 - A small smoke trail
- Flames should disappear quickly.
- Repeated taps may briefly create a larger visual flame, but the effect should have a maximum size.

The fire effect should not create permanent burn marks unless a separate premium variant is introduced later.

Potential future variants:

- Purple fire
- Blue fire
- Ghost fire
- Pixel fire
- Golden fire

6. Autumn Leaves Tap Effect

Autumn leaves appear at the tap location and fall toward the ground.

- Price: **200 Hero Coins**
- Leaves should vary between:
 - Red
 - Orange
 - Yellow
 - Brown
 - Different leaf shapes
 - Different falling rotations

Leaves accumulate into temporary piles when they reach the ground.

Leaf Pile Interaction

If the player lands in or moves through a large leaf pile:

- Leaves burst outward across the screen.
- The character may briefly disappear into the pile.
- A landing or jumping-in-leaves animation plays.
- Leaf particles scatter in multiple directions.
- Leaves settle again or disappear after the interaction.

The pile should gradually fade or blow away after a configured duration.

This effect is particularly suitable for Infinity Taps because players can intentionally build piles and repeatedly jump into them.

Store Summary

| Tap Effect Price |
|---|
| --- ---: |
| Purple Pulsing Rings 100 Coins |
| Additional Custom Ring Color 50 Coins |
| RGB Pulsing Rings 300 Coins |
| Rain and Rare Lightning 200 Coins |
| Snow and Evil Snowman 200 Coins |
| Magma and Obsidian Golem 200 Coins |
| Fire 75 Coins |
| Autumn Leaves 200 Coins |

The fire effect acts as an affordable early purchase. The RGB rings provide a higher-priced prestige option, while rain, snow, magma, and leaves provide interactive environmental effects.

System Architecture

Recommended Godot structure:

```
```text
TapEffectManager
TapEffectDefinition
TapEffectInstance
```

```
TemporaryEnvironmentManager
RareEffectManager
CosmeticLoadoutManager
PerformanceSettings
...
```

### ## Tap Effect Manager

The manager listens for valid tap events:

```
``gdscript
signal tap_registered(
 screen_position: Vector2,
 world_position: Vector2
)
...
```

It then:

1. Checks the equipped effect.
2. Retrieves its configuration.
3. Spawns the visual effect.
4. Applies environment-specific behavior.
5. Processes rare-event eligibility.
6. Enforces particle and object limits.

Gameplay scripts should not contain separate logic for every cosmetic.

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### # Tap Effect Definition

Each effect should be data-driven.

```
``json
{
 "id": "tap_effect_rain",
 "display_name": "Storm Touch",
 "price": 200,
 "effect_scene": "res://effects/tap/rain_effect.tscn",
 "supports_custom_color": false,
 "supports_accumulation": true,
 "rare_event_id": "lightning_strike",
 "rare_event_chance": 0.00001,
}
```

```
"max_active_instances": 8,
"duration_seconds": 4.0,
"enabled": true
}
...
```

A probability of `0.00001` represents **0.001%**.

---

## # Accumulation System

Snow, lava, and leaves require a temporary environmental accumulation system.

Avoid creating hundreds of permanent physics objects. Instead, divide terrain into accumulation zones.

Example:

```
``text
Zone ID
Effect Type
Current Amount
Maximum Amount
Last Updated Time
Current Visual Stage
Rare Spawn Checked
``
```

When a particle reaches the ground, increase the zone's amount rather than leaving every particle active indefinitely.

Example logic:

```
``gdscript
func add_accumulation(
 zone_id: StringName,
 effect_type: StringName,
 amount: float
) -> void:
 var zone := get_or_create_zone(zone_id, effect_type)
 zone.current_amount = min(
 zone.current_amount + amount,
 zone.maximum_amount
```

```
)
 update_zone_visual(zone)
...
```

This allows the visual pile to grow without creating excessive processing cost.

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## # Rare Character Logic

The Evil Snowman and Obsidian Golem should use threshold-based checks.

Example:

```
```text  
Snow pile reaches 75% capacity:  
Check snowman spawn chance once.
```

```
Lava pool reaches 75% capacity:  
Check golem spawn chance once.  
```
```

Do not perform a rare-character roll for every particle.

Possible default chances:

- Evil Snowman: 3–5% once a qualifying pile forms
- Obsidian Golem: 3–5% once a qualifying pool forms

These values can be adjusted after testing.

A failed roll may reset once the pile fully melts or disappears, allowing another future qualifying pile to receive a new check.

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## # Performance Requirements

Tap effects must remain stable during rapid tapping.

Set limits for:

- Maximum visible particles
- Maximum simultaneous rain emitters

- Maximum accumulation zones
- Maximum active leaf piles
- Maximum temporary creatures
- Maximum ground decals
- Effect lifetime
- Object pooling

Recommended approach:

- Use `GPUParticles2D` where supported.
- Provide a `CPUParticles2D` or simplified fallback where necessary.
- Pool repeating objects rather than repeatedly creating and deleting them.
- Merge nearby accumulation zones.
- Reduce effects automatically on lower-end devices.

Suggested quality settings:

```
```text
Full
Reduced
Minimal
Off
```
```

At reduced quality:

- Fewer rain drops
- Fewer snowflakes
- Smaller leaf bursts
- Simplified lava flow
- No persistent burn decals
- Reduced rare-character particles

Rare events should remain available even when effect quality is reduced.

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## # Multiplayer Behavior

Tap effects may be shown in online play, but they must not obstruct other players.

Online rules:

- Other players see a reduced version of the effect.

- Local player sees the full equipped effect.
- Persistent snow, lava, and leaf piles should remain client-side.
- Rare creatures should normally be visible only to the owner unless synchronized as a lightweight cosmetic event.
- Effects must not change collision, visibility, or input.
- Players should have a setting to reduce or hide other players' effects.

Recommended setting:

```
```text
```

Other Player Effects:

Full | Reduced | Off

```
```
```

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## # Accessibility

Provide controls for:

- Particle density
- Effect opacity
- Screen shake
- Flash intensity
- Lightning flash
- Cosmetic sound volume
- Haptic feedback
- Other-player effects

The lightning event should not use a pure full-screen white flash. Use a short controlled brightness increase with an accessibility-safe alternative.

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## # Ownership and Customization Data

Store:

```
```text
```

owned_tap_effect_ids

equipped_tap_effect_id

tap_effect_settings

```
```
```



Example:

```
```json
{
  "equipped_tap_effect_id": "tap_effect_pulse_ring",
  "tap_effect_settings": {
    "tap_effect_pulse_ring": {
      "mode": "custom_color",
      "color": "#7B35D4",
      "opacity": 0.85
    }
  }
}
```
```

For the RGB version:

```
```json
{
  "mode": "smooth_rainbow",
  "speed": 1.0
}
```
```

---

### # Important Design Rule

These tap effects may create visual interactions and rare cosmetic characters, but none should:

- Award additional coins
- Increase score
- Improve survival time
- Change gravity
- Affect jump force
- Damage players
- Obstruct opponents
- Alter rare-item discovery odds
- Provide achievement advantages unavailable to free players

They are purchased because they make tapping more expressive, satisfying, and visually distinctive.

# Additional Tap Effect Ideas

The strongest tap effects should do more than produce a brief particle burst. They should create small environmental reactions, collectible moments, or satisfying chain effects—especially in Infinity Taps.

## 1. Bubble Pop

A cluster of translucent bubbles appears around the tap point. Bubbles float upward and pop after a few seconds.

- Larger bubbles may contain tiny stars, fish, or faces.
- Tapping an existing bubble pops it immediately.
- Rare chance for a giant bubble to carry the character briefly.
- Suggested cost: **100 Coins**

## 2. Pixel Lightning

A small electrical bolt jumps from the tap point to nearby surfaces.

- Repeated taps build static electricity.
- At maximum charge, several bolts chain across the screen.
- Rare chance to temporarily create a small electric creature.
- Suggested cost: **175 Coins**

This should remain separate from the rare lightning event within the rain effect by looking more arcade-like and electrical rather than weather-based.

## 3. Starfall

Each tap creates a small falling star that strikes the location.

- Stars leave temporary glowing marks.
- Consecutive taps form a constellation.
- Completing certain constellation patterns could trigger a cosmetic animation.
- Suggested cost: **200 Coins**

## 4. Galaxy Portal

A tiny cosmic portal opens at the tap location.

- Stars and particles spiral inward.
- The portal closes after a brief period.
- Rare chance for an alien hand, eye, or creature to peek through.
- Suggested cost: **250 Coins**

## 5. Ghost Hands

Translucent ghostly paws or hands briefly emerge where the player taps.

- They may wave, clap, point, or try to grab the character.
- Several nearby taps could summon a temporary ghost audience.
- Suggested cost: **175 Coins**

This would pair naturally with Velvet Hex.

## 6. Slime Splash

A colorful slime splash appears and temporarily sticks to the environment.

- Repeated taps create larger slime puddles.
- The character bounces or splashes visually when landing in them.
- Rare chance for a tiny slime creature to emerge.
- Suggested cost: **150 Coins**

The bounce must remain visual and must not change actual jump force.

## 7. Ink Blot

Dark ink splashes outward from the tap point.

- Each splash forms a random shape.
- Blots slowly fade or drip downward.
- Rare shapes may resemble creatures, faces, or symbols.
- Suggested cost: **125 Coins**

## 8. Paint Burst

Each tap splashes paint onto the environment.

- Players choose a single color, random colors, or a custom palette.
- Multiple taps can temporarily paint sections of the ground.
- Colors slowly fade back to normal.
- Suggested cost:
  - Base color: **100 Coins**
  - Custom palette: **200 Coins**
  - Rainbow paint: **300 Coins**

## 9. Flower Bloom

Flowers rapidly grow from the tap location.

- Different taps produce different flowers.
- Repeated tapping creates a temporary garden.
- Butterflies may appear around large flower patches.
- Rare chance for a giant strange flower to bloom.
- Suggested cost: **200 Coins**

## 10. Mushroom Growth

Small mushrooms sprout wherever the player taps.

- Mushrooms vary in size, color, and shape.
- Larger clusters may release spores.
- Rare chance for a mushroom creature to wake up and walk away.
- Suggested cost: **200 Coins**

## 11. Crystal Growth

Crystals rise from the ground beneath the tap point.

- Repeated taps make the cluster larger.
- Crystals shimmer and eventually dissolve.
- Rare chance for a crystal guardian or tiny crystal animal to appear.
- Suggested cost: **225 Coins**

## 12. Mini Earthquake

A small crack forms beneath the tap point.

- Dust and tiny stones bounce upward.
- Repeated taps extend the crack visually.
- A large chain may cause a harmless rock creature to peek out.
- Suggested cost: **175 Coins**

Screen shake should be optional and disabled by default for accessibility.

## 13. Water Splash

A water droplet falls and splashes at the tap point.

- Repeated taps create temporary puddles.
- Landing in a puddle creates a larger splash.
- Puddles may reflect the character or environment.
- Rare chance for a frog or tiny water creature to appear.
- Suggested cost: **150 Coins**

## 14. Ocean Wave

A small wave sweeps outward from the tap location.

- Consecutive taps make waves slightly larger.
- Sea foam remains briefly.
- Rare chance for a tiny boat, fish, or message bottle to ride the wave.
- Suggested cost: **225 Coins**

## 15. Sand Burst

Sand erupts and falls around the tap point.

- Repeated taps create temporary sand piles.
- The character can visually sink into or scatter them.
- Rare chance for a scarab, buried object, or miniature sand creature to emerge.
- Suggested cost: **175 Coins**

## 16. Candy Explosion

Candy pieces burst from each tap.

- Includes wrapped candy, gumdrops, jellybeans, and lollipops.
- Candy bounces briefly before fading.
- Rare chance for a candy monster or piñata-like creature to appear.
- Suggested cost: **200 Coins**

## 17. Confetti Cannon

Confetti bursts outward from the tap point.

- Different themes could include party, birthday, victory, or metallic confetti.
- Sustained tapping creates a larger celebration.
- Suggested cost: **100 Coins**

This would be a strong affordable cosmetic because it is simple, cheerful, and performance-friendly.

## 18. Musical Notes

Each tap produces a visual musical note and a soft tone.

- Tap position determines pitch.
- Fast tapping creates simple melodies.
- Players can choose instrument sets such as:

- Piano
- Chiptune
- Bells
- Drums
- Slime pops
- Suggested cost: **150 Coins per sound set**

This could become one of the strongest Infinity Taps effects because it directly enhances fidgeting.

## 19. Retro Arcade Hit

Each tap creates an old arcade-style impact effect.

Possible visuals:

- POW!
- TAP!
- Pixel stars
- Score numbers
- CRT flicker
- Comic-style bursts

Suggested cost: **125 Coins**

A higher-tier version could randomize retro words and symbols.

## 20. Glitch Effect

The tapped area briefly distorts into pixel fragments.

- Colors split into RGB channels.
- Small blocks shift out of position.
- Repeated taps create temporary digital corruption.
- Rare chance for a glitch character to appear.
- Suggested cost: **250 Coins**

The effect should be localized and should not distort the entire screen.

## 21. Shadow Creatures

A small shadow briefly emerges from the tap location.

- It may crawl, wave, run away, or hide.
- Several taps can create a temporary group of shadow creatures.

- Rare chance for one to become a named encounter.
- Suggested cost: **225 Coins**

## 22. Tiny Footprints

Each tap leaves a temporary footprint.

Possible variants:

- Cat paws
- Demon footprints
- Slime marks
- Bird feet
- Dinosaur tracks
- Alien prints

Repeated taps create a visible trail through the world.

Suggested cost:

- Individual footprint type: **75 Coins**
- Footprint bundle: **250 Coins**

## 23. Magic Runes

A glowing rune appears under each tap.

- Runes rotate and fade.
- Several nearby runes may connect into a larger magic circle.
- Completing a circle could summon a harmless magical creature or visual portal.
- Suggested cost: **225 Coins**

## 24. Black Hole

A tiny black hole forms at the tap point.

- Nearby cosmetic particles are pulled toward it.
- It closes after a short duration.
- Repeated taps strengthen the visual pull.
- Rare chance for something strange to be expelled when it closes.
- Suggested cost: **300 Coins**

The pull must affect only cosmetic particles, never the player or gameplay objects.

## 25. Cloud Puff

Each tap creates a small cloud.

- Clouds drift slowly.
- Repeated taps create larger temporary cloud formations.
- Rare chance for a face, animal shape, or tiny storm cloud to form.
- Suggested cost: **125 Coins**

## 26. Fireworks

Each tap launches a small firework that explodes shortly afterward.

- Different shapes include circles, stars, hearts, cat faces, and spirals.
- Repeated tapping creates a coordinated display.
- Suggested cost:
  - Basic fireworks: **200 Coins**
  - Custom colors: **250 Coins**
  - Shape bundle: **350 Coins**

## 27. Miniature Meteor

A small meteor falls toward the tap point.

- Creates a temporary crater and smoke burst.
- Repeated taps create a small meteor shower.
- Rare chance for a tiny alien object or cosmic creature to emerge.
- Suggested cost: **225 Coins**

## 28. Hearts and Affection

Hearts burst from the tap point and float upward.

- Different heart styles could include pixel, gothic, broken, slime, or rainbow.
- Nearby characters may react positively.
- Suggested cost: **75 Coins**

## 29. Bat Swarm

A few tiny bats fly away from the tap location.

- Consecutive taps create a larger swarm.
- Rare chance for a larger named bat companion to appear briefly.
- Suggested cost: **175 Coins**

This would fit naturally with Velvet Hex and seasonal Halloween content.



### 30. Tiny Companion Reaction

Instead of a standard particle, a miniature companion appears at the tap location.

Examples:

- Tiny slime
- Mini ghost
- Frog
- Alien
- Snow creature
- Fire sprite

The companion performs one quick animation and disappears.

Suggested cost: **250–400 Coins**, depending on complexity.

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## Strongest Additions for the First Version

I would prioritize these first because they offer distinct visual identities without excessive development cost:

| Effect           | Suggested Price | Why it works                           |
|------------------|-----------------|----------------------------------------|
| Bubble Pop       | 100             | Relaxing and interactive               |
| Slime Splash     | 150             | Fits Velvet Hex and Infinity Taps      |
| Musical Notes    | 150             | Enhances fidgeting directly            |
| Flower Bloom     | 200             | Creates temporary environments         |
| Retro Arcade Hit | 125             | Supports the game’s pixel-art identity |
| Magic Runes      | 225             | Allows satisfying combination effects  |
| Galaxy Portal    | 250             | Supports rare encounters               |
| Tiny Footprints  | 75              | Affordable entry cosmetic              |
| Confetti Cannon  | 100             | Simple and cheerful                    |
| Glitch Effect    | 250             | Strong premium visual identity         |

## Pricing Structure Recommendation

To keep the store understandable:

- **75–100 Coins:** Simple effects
- **125–175 Coins:** Effects with additional movement or variation
- **200–250 Coins:** Accumulating effects or rare-event interactions
- **300–400 Coins:** Highly customizable or elaborate premium effects

Not every effect needs a hidden creature or rare event. If every effect summons something, those discoveries will stop feeling special. Approximately one-third of tap effects should have deeper accumulation or rare-event behavior, while the others remain immediate, satisfying cosmetic responses.

# Player Particle Effects

# Tap Against Me — Character Particle Effects

## Developer Handoff

Character particle effects are cosmetics attached directly to the player character. Unlike tap effects, which appear where the screen is tapped, character particles follow the player while moving, falling, jumping, floating, swimming, or remaining idle.

They must be entirely cosmetic and must never alter:

- Jump force
- Gravity
- Movement speed
- Collision size
- Fruit collection range
- Coin earnings
- Rare-spawn chances
- Multiplayer visibility advantages

Players may equip **one primary character particle effect** at a time.

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## 1. Particle Effect Behavior

A character particle effect may include several components:

- **Movement trail:** Appears behind the player while moving.
- **Idle particles:** Appear while the character remains mostly still.
- **Jump burst:** Triggers when upward movement begins.
- **Falling trail:** Changes while the character is descending.
- **Landing burst:** Plays when the character contacts the ground in Infinity Taps.
- **Environment variation:** Changes appearance underwater, in space, or inside special biomes.

Not every particle effect needs all components. Simpler effects can use only a movement trail, while premium effects can include a complete visual package.

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## 2. Suggested Character Particle Effects

### Purple Slime Droplets

Small purple slime droplets separate from the character and fall behind it.

- Droplets stretch while falling.
- Some droplets splatter briefly on the ground.
- Underwater, they become floating slime bubbles.
- Strong thematic match for Velvet Hex.
- Suggested cost: **100 Hero Coins**

### Shadow Smoke

Dark smoke trails behind the character.

- Smoke curls and slowly fades.
- Purple highlights appear inside the smoke.
- While idle, small shadow wisps orbit the character.
- Suggested cost: **150 Hero Coins**

### Golden Sparkles

Small golden stars and sparkles trail behind the player.

- Produces a short golden burst when jumping.
- Sparkles linger briefly at the highest point of a jump.
- Suggested cost: **150 Hero Coins**

### Poison Bubbles

Purple and green toxic bubbles rise from the character.

- Bubbles vary in size.
- Occasionally, a bubble contains a tiny skull symbol.
- Bubbles pop without affecting gameplay.
- Suggested cost: **125 Hero Coins**

## Pixel Stars

Retro arcade stars trail behind the character.

- Uses a limited 1990s arcade palette.
- Stars switch between several blocky shapes.
- Jumping creates a brief starburst.
- Suggested cost: **100 Hero Coins**

## Fire Trail

Small flames and embers stream behind the character.

- Flames become stronger while falling quickly.
- Idle particles appear as small sparks.
- Underwater, flames become steam bubbles or are visually reduced.
- Suggested cost: **175 Hero Coins**

## Blue Ghost Fire

Small blue-purple flames orbit and trail behind the player.

- Flames resemble harmless spirits.
- While idle, one or two flames hover nearby.
- Strong match for supernatural characters.
- Suggested cost: **200 Hero Coins**

## Snowflake Trail

Snowflakes fall away from the character as it moves.

- Faster movement creates denser snow.
- Idle state produces a light frost mist.
- Landing creates a small snow puff.
- Suggested cost: **150 Hero Coins**

## Autumn Leaf Trail

Leaves stream behind the character.

- Includes red, orange, yellow, and brown leaves.
- Leaves rotate and drift differently.

- Landing produces a short leaf burst.
- Suggested cost: **150 Hero Coins**

## Flower Petals

Soft petals trail behind the player.

- Possible variants:
  - Cherry blossom
  - Rose petals
  - Sunflower petals
  - Purple fantasy flowers
- Petals drift more slowly than standard particles.
- Suggested cost: **150 Hero Coins per theme**

## Bubble Trail

Translucent bubbles follow the character.

- Bubbles float upward after spawning.
- Underwater, the trail becomes denser and more natural.
- A few bubbles may pop when the player changes direction.
- Suggested cost: **125 Hero Coins**

## Ocean Foam

Small water droplets and white foam trail behind the character.

- On land, it appears as splashing water.
- Underwater, it becomes a stream of tiny bubbles.
- Suggested cost: **150 Hero Coins**

## Lightning Static

Electric sparks jump around the character.

- Faster movement increases the electrical activity.
- Jumping creates a small lightning burst beneath the player.
- No full-screen flashes.
- Suggested cost: **200 Hero Coins**

## Rainbow Trail

A short rainbow-colored trail follows the character.

- The trail should fade quickly to prevent screen clutter.
- Optional modes:
  - Smooth spectrum
  - Pastel rainbow
  - Retro arcade rainbow
- Suggested cost: **300 Hero Coins**

## RGB Pixel Trail

Small square pixels rapidly change colors behind the character.

- Different from the smooth Rainbow Trail.
- Designed to resemble digital arcade particles.
- Supports random, cycling, or custom palette modes.
- Suggested cost: **300 Hero Coins**

## Glitch Fragments

Small pieces of the character appear to break into digital fragments.

- Pixel blocks shift and dissolve behind the character.
- Occasional RGB color separation appears near the sprite.
- Must remain localized around the player.
- Suggested cost: **250 Hero Coins**

## Cosmic Dust

Stars, tiny planets, and purple cosmic dust trail behind the player.

- In space, the effect becomes slightly more visible.
- Idle particles gently orbit the character.
- Suggested cost: **250 Hero Coins**

## Mini Comets

Tiny comets follow the character's movement.

- Each has a short glowing tail.
- During a fast fall, several comets appear together.
- Suggested cost: **225 Hero Coins**



## Black Hole Orbit

Tiny dark particles orbit the character and are absorbed into a miniature central void.

- The void should remain visually small.
- Nearby cosmetic particles may bend slightly toward it.
- Must not pull gameplay objects.
- Suggested cost: **300 Hero Coins**

## Tiny Bats

Small pixel bats periodically fly from behind the character.

- Bats briefly follow before scattering.
- Idle state may show one bat hanging upside down nearby.
- Suggested cost: **200 Hero Coins**

## Ghost Companions

Tiny ghost particles follow the character.

- Ghosts make different faces.
- They may circle, wave, or disappear into smoke.
- Suggested cost: **250 Hero Coins**

## Tiny Slimes

Miniature slime creatures bounce behind the player.

- They appear briefly rather than remaining permanent.
- Colors may match the equipped character skin.
- Suggested cost: **250 Hero Coins**

## Cat Paw Prints

Temporary paw prints appear along the character's movement path.

- On the ground, prints appear directly on the surface.
- In the air, stylized translucent paw symbols trail behind.
- Suggested cost: **100 Hero Coins**

## Demon Footprints

Purple burning footprints briefly appear behind the player.

- Includes small horns or spade-tail symbols.
- Marks fade after a few seconds.
- Suggested cost: **125 Hero Coins**

## Hearts

Small hearts float away from the character.

- Available styles:
  - Standard red
  - Purple gothic
  - Pixel hearts
  - Broken hearts
  - Rainbow hearts
- Suggested cost: **75–125 Hero Coins**

## Musical Notes

Pixel musical notes trail behind the character.

- Notes may pulse in time with tap sounds.
- Optional sound packs can be selected separately.
- Suggested cost: **150 Hero Coins**

## Confetti

Small pieces of confetti burst behind the character while moving.

- Faster movement creates slightly larger bursts.
- Best used with strict particle limits.
- Suggested cost: **125 Hero Coins**

## Candy Trail

Candy pieces, jellybeans, and wrapped sweets trail behind the player.

- Items bounce briefly before fading.
- Suggested cost: **175 Hero Coins**

## Crystal Shards

Tiny crystal fragments trail behind the character.

- Colors can include purple, blue, red, green, or rainbow.
- Shards briefly sparkle before dissolving.
- Suggested cost: **175 Hero Coins**

## Mushroom Spores

Soft glowing spores follow the player.

- Occasional tiny mushroom icons appear.
- In forests, the effect may become slightly more visible.
- Suggested cost: **175 Hero Coins**

## Fireflies

Small glowing fireflies surround and follow the character.

- They move independently rather than forming a straight trail.
- While idle, they slowly orbit.
- Suggested cost: **175 Hero Coins**

## Cloud Trail

Small cloud puffs form behind the player.

- During rapid upward movement, the trail resembles a cartoon rocket launch.
- Rare cloud puffs may briefly form recognizable shapes.
- Suggested cost: **150 Hero Coins**

## Rain Cloud Companion

A tiny cloud follows above the player and produces rain.

- The cloud reacts to movement with a short delay.
- Rain remains cosmetic and does not accumulate.
- Suggested cost: **250 Hero Coins**

## Halo Light

Soft rays and small light particles surround the character.

- A pixel halo briefly appears during jump peaks.
- Suggested cost: **225 Hero Coins**

## Dark Aura

A subtle purple-black aura surrounds the character.

- Small particles rise from the body.
- The outline pulses gently.
- Must not make the player difficult to see.
- Suggested cost: **200 Hero Coins**

## Crown Sparkles

A floating crown-shaped sparkle occasionally appears above the player.

- Smaller golden particles follow behind.
- Suitable for ranked or achievement rewards.
- Suggested cost: **250 Hero Coins**, or achievement-exclusive

## Retro Score Numbers

Small arcade score values appear behind the character.

Examples:

+10  
+25  
TAP!  
NICE!  
WOW!

These numbers are cosmetic and do not represent real score changes.

- Suggested cost: **150 Hero Coins**

## CRT Afterimage

The character leaves brief retro-colored afterimages behind it.

- Afterimages use limited cyan, magenta, and purple tones.
- Designed to resemble an old arcade display.

- Suggested cost: **225 Hero Coins**

## Matrix Code

Small pixel symbols and numbers fall behind the player.

- Uses fictional or abstract characters rather than readable messages.
- Suggested cost: **200 Hero Coins**

## Floating Runes

Small magical symbols orbit and trail behind the player.

- Jumping briefly connects them into a circular pattern.
- Suggested cost: **225 Hero Coins**

## Moon Dust

Soft gray and purple dust trails behind the character.

- Particularly visible on the Moon and other rocky worlds.
- Landing creates a wider dust puff.
- Suggested cost: **125 Hero Coins**

## Alien Spores

Bright green and purple particles float around the player.

- Some spores briefly form alien-eye shapes.
  - Suitable as a space exploration reward.
  - Suggested cost: **200 Hero Coins**
- 

# 3. Premium Complete Effect Sets

Certain higher-priced effects may combine trails, idle particles, jump bursts, and landing effects.

## Velvet Hex Shadow Set

- Purple slime movement droplets
- Shadow-smoke idle aura
- Purple rune jump burst
- Slime landing splash
- Suggested cost: **400 Hero Coins**

## Cosmic Traveler Set

- Cosmic dust trail
- Orbiting stars while idle
- Comet jump burst
- Miniature black-hole landing effect
- Suggested cost: **500 Hero Coins**

## Frozen Spirit Set

- Snowflake trail
- Frost aura
- Ice crystal jump burst
- Snow landing puff
- Suggested cost: **400 Hero Coins**

## Infernal Set

- Fire trail
- Ember aura
- Flame jump burst
- Magma landing splash
- Suggested cost: **400 Hero Coins**

## Retro Arcade Set

- RGB pixel trail
- CRT afterimage
- Pixel star jump burst
- **TAP!** landing text
- Suggested cost: **500 Hero Coins**

These complete sets should still occupy one character-particle slot.

---

## 4. Seasonal Particle Effects

Seasonal passes should include unique character particles that cannot be purchased after the season ends.

Examples:

| Season         | Particle effect                              |
|----------------|----------------------------------------------|
| Halloween      | Ghosts, bats, or cursed purple flames        |
| Winter         | Snowflakes, frost mist, or tiny snow spirits |
| Spring         | Flowers, petals, and butterflies             |
| Summer         | Sun sparkles, water droplets, or fireflies   |
| Space Season   | Cosmic dust, stars, and alien spores         |
| Arcade Season  | RGB pixels, scanlines, and score numbers     |
| Volcano Season | Embers, ash, and lava droplets               |
| Ocean Season   | Bubbles, foam, and tiny fish                 |

Seasonal effects remain permanently owned once unlocked.

---

## 5. Character-Specific Effects

Some particle effects may be designed for one character but should not provide a competitive advantage.

For Velvet Hex:

- Shadow Slime
- Purple Curse Smoke
- Tiny Bats
- Stitched Ghosts
- Hex Runes
- Toxic Bubbles

- Purple Paw Prints

Character-specific particles can be sold individually or included in character bundles.

A character-specific effect should clearly state compatibility in the shop:

Compatible with Velvet Hex and Velvet Hex skins.

Universal effects should work with every character.

---

## 6. Pricing Structure

| Effect complexity                   | Suggested price |
|-------------------------------------|-----------------|
| Simple trail                        | 75–125 Coins    |
| Trail with movement variation       | 125–175 Coins   |
| Trail with idle and landing effects | 175–250 Coins   |
| Premium animated effect             | 250–350 Coins   |
| Complete particle set               | 400–500 Coins   |
| Seasonal exclusive                  | Season Pass     |
| Achievement effect                  | Not purchasable |

Players should also receive some free particle effects through achievements, exploration, and seasonal free tracks so the system does not feel exclusively paid.

---

## 7. Particle Loadout

The cosmetic loadout should include:

Character  
Character Skin  
Character Particle Effect  
Tap Effect



## Profile Badge

Example:

```
{
 "character_id": "velvet_hex",
 "skin_id": "velvet_hex_toxic",
 "particle_effect_id": "particle_shadow_slime",
 "tap_effect_id": "tap_effect_purple_rings",
 "primary_badge_id": "badge_poison_fruit"
}
```

The shop preview should allow the player to:

- Move the character left and right
  - Simulate a jump
  - Simulate falling
  - View the idle effect
  - View the landing effect
  - Compare full and reduced particle settings
- 

## 8. Godot Implementation Notes

Recommended structure:

CharacterParticleManager  
CharacterParticleDefinition  
CharacterParticleController  
ParticleObjectPool  
CosmeticLoadoutManager  
PerformanceSettings

Each character should contain attachment markers:

ParticleAnchorCenter  
ParticleAnchorBack  
ParticleAnchorFeet  
ParticleAnchorHead  
ParticleAnchorTail

Different particle components should attach to the appropriate marker.

Example:

- Trails: `ParticleAnchorBack`
- Auras: `ParticleAnchorCenter`
- Crowns and clouds: `ParticleAnchorHead`
- Landing effects: `ParticleAnchorFeet`
- Tail-specific effects: `ParticleAnchorTail`

## Particle Definition

```
{
 "id": "particle_shadow_slime",
 "display_name": "Shadow Slime",
 "price": 150,
 "universal": true,
 "compatible_characters": [],
 "trail_scene": "res://effects/character/shadow_slime_trail.tscn",
 "idle_scene": "res://effects/character/shadow_slime_idle.tscn",
 "jump_scene": "res://effects/character/shadow_slime_jump.tscn",
 "landing_scene": "res://effects/character/shadow_slime_land.tscn",
 "quality_level": "medium",
 "enabled": true
}
```

Character movement scripts should emit general events:

```
signal character_jump_started
signal character_falling_started
signal character_landed
signal character_idle_started
signal character_movement_changed(velocity: Vector2)
```

The particle manager responds to these signals. Character movement code should not reference individual cosmetic IDs.

---

## 9. Performance Rules

Character effects will be active much more often than tap effects, so stricter limits are required.

Use:

- Object pooling
- Fixed particle lifetimes
- Maximum particle counts
- Emission-rate limits
- Reduced off-screen processing
- Simplified multiplayer versions
- Lower-density settings for older devices

Recommended quality options:

Full  
Reduced  
Minimal  
Off

At reduced quality:

- Lower emission rate
- Shorter trail length
- Fewer orbiting objects
- Simplified landing burst
- No secondary particles

At minimal quality:

- One basic trail component only
- No idle aura
- No environment variation

---

## 10. Multiplayer Visibility

The local player sees the full effect.

Other players should see:

- Reduced particle count
- Shorter trail duration
- No large persistent particles
- No screen distortion
- No cosmetic audio unless explicitly enabled

Player setting:

Other Player Particles:

Full | Reduced | Off

Particle effects must never hide a player's collision position or make one skin harder to track than another.

---

## 11. Recommended Initial Set

For the first release, I would build:

1. Purple Slime Droplets
2. Shadow Smoke
3. Golden Sparkles
4. Pixel Stars
5. Fire Trail
6. Snowflake Trail
7. Bubble Trail
8. Autumn Leaves
9. Tiny Bats
10. Rainbow Trail
11. Glitch Fragments
12. Cosmic Dust

This gives the shop a mixture of affordable, thematic, premium, and seasonal-quality options without requiring dozens of highly complex systems at launch.

# Infinity Taps

## # Tap Against Me — Infinity Taps Mode

### ## Developer Handoff

#### ### 1. Mode Overview

**\*\*Infinity Taps\*\*** is a noncompetitive local game mode designed for casual play, fidgeting, relaxation, and open-ended exploration. There is no losing, no ground collision failure, no increasing gravity challenge, and no pressure to achieve a high score.

The player begins on the ground in a basic natural environment. Every tap moves the character through the world while permanently increasing the player's total tap count. By tapping in different areas of the screen, the player can move left, right, upward, downward where permitted, or continue exploring in the current direction.

Over time, the player can travel through fields, forests, deserts, oceans, the upper atmosphere, space, moons, planets, and other unusual environments. Rare characters, creatures, events, and cosmetics can appear during exploration.

The intended experience is:

> **\*\*Tap, move, discover, and relax without ever losing.\*\***

---

#### # 2. Design Goals

Infinity Taps should provide:

- A stress-free fidgeting activity
- Permanent tap-count progression
- A sense of traveling through a large connected universe
- Rare discoveries that make long sessions memorable
- Light cosmetic rewards without flooding the coin economy
- A place to enjoy owned characters, skins, trails, and tap effects
- Reasons to revisit previously discovered areas
- Casual play without rankings, PvP, timers, or failure conditions

The mode should not require careful timing or mechanical mastery. A player should be able to tap absentmindedly while listening to music, watching something, resting, or using the game as a fidget activity.

---

### # 3. Core Gameplay Loop

The core loop is:

1. Enter Infinity Taps.
2. Begin at the player's last saved location.
3. Tap anywhere on the screen.
4. Increase the permanent total tap count.
5. Move according to tap location.
6. Travel through environments.
7. Encounter creatures, characters, landmarks, and rare events.
8. Occasionally discover cosmetics or collectible entries.
9. Earn one Hero Coin for every 1,000 valid taps.
10. Continue indefinitely or exit at any time.

There is no round start or round end. The mode is one persistent exploration session that continues across multiple visits.

---

### # 4. Tap Movement

The screen should be divided into broad directional regions.

#### ## Basic Controls

- Tap left of the character to move left.
- Tap right of the character to move right.
- Tap above the character to move upward or jump.
- Tap below the character to descend when swimming, flying, or in space.
- Tap near the character to produce a smaller bounce or mostly remain in place.

The player should not need an on-screen joystick.

Each valid tap should:

- Increment the tap counter by one
- Trigger the equipped tap effect
- Play a small character animation
- Apply directional movement
- Produce appropriate particles or environmental feedback

Movement should remain forgiving. Repeated tapping should feel rhythmic rather than physically difficult.

### ## Contextual Movement

Movement changes based on the environment:

| Environment | Movement behavior |  
|---|---|  
| Ground areas | Bounce, hop, and travel left or right |  
| Mountains | Climb upward through repeated taps |  
| Sky | Float and rise between clouds |  
| Space | Drift freely in all directions |  
| Water surface | Swim horizontally or dive |  
| Deep ocean | Free directional swimming |  
| Low-gravity moons | Longer, slower jumps |  
| High-gravity planets | Shorter, heavier bounces |  
| Ice regions | Slight sliding after movement |  
| Slime or alien worlds | Elastic or unusual movement |

These differences should be primarily visual and experiential. They should not create failure conditions.

---

### # 5. No-Loss Rules

Infinity Taps must not contain a traditional loss state.

The player cannot:

- Die
- Run out of lives
- Hit the ground and lose
- Fail an exploration session
- Lose coins or discoveries
- Be eliminated by creatures
- Be forced to restart from the beginning

Environmental hazards can still exist visually, but they should create humorous or temporary reactions instead of punishment.

Examples:



- Lava launches the character upward.
- A whale blows the character toward the surface.
- A meteor sends the character spinning.
- An alien tractor beam relocates the player.
- A strong ocean current carries the player elsewhere.
- A sandstorm briefly obscures the environment.

These events should make exploration more interesting without threatening progress.

---

## # 6. Tap Counter and Coin Rewards

Infinity Taps has its own persistent tap counter.

Track:

- Current session taps
- Lifetime Infinity Taps
- Total taps across the entire game
- Taps earned in each biome
- Highest single-session tap count
- Milestone progress

## ## Coin Rule

For every **\*\*1,000 valid Infinity Taps\*\***, the player receives:

> **\*\*1 Hero Coin\*\***

Coins do not physically spawn in this mode.

Examples:

| Lifetime eligible taps   Total Hero Coins earned |
|--------------------------------------------------|
| ---: ---:                                        |
| 1,000   1                                        |
| 10,000   10                                      |
| 100,000   100                                    |
| 1,000,000   1,000                                |

The system should preserve remainder progress.

Example:

```
```text
```

Current eligible taps: 7,450

Coins already awarded: 7

Progress toward next coin: 450 / 1,000

```
```
```

Do not reset the remainder when the player closes the game.

## ## Reward Presentation

Every 1,000 taps, show a small nonintrusive notification:

> 1,000 Taps Reached

> +1 Hero Coin

The notification should not interrupt tapping or stop movement.

Milestones such as 10,000, 100,000, and 1,000,000 taps may receive larger celebrations, badges, or cosmetic rewards.

---

## # 7. Coin-Economy Safeguards

Because Infinity Taps awards spendable currency, it cannot rely entirely on an unrestricted local counter.

The developer should account for:

- Auto-clickers
- Macros
- Modified save files
- Artificial touch injection
- Extreme input rates
- Device clock manipulation
- Repeated offline-save duplication

## ## Valid Tap Rules

A tap should count only when:

- The application is active and visible.

- The input comes through an approved touch or mouse event.
- The tap interval falls within reasonable human limits.
- The mode is not paused or covered by a menu.
- The same input event has not already been processed.

Do not hard-limit normal fast tapping. Use a generous threshold designed only to reject clearly impossible input.

For example, inputs sustained above 15–20 taps per second for an extended period could be flagged rather than immediately punished.

### ## Offline Currency

The mode can operate offline, but coin rewards should be marked as locally pending until account synchronization.

Recommended behavior:

1. Count taps locally.
2. Award pending Hero Coins at each 1,000-tap threshold.
3. Save a signed or checksummed transaction record.
4. Upload the summarized session when connected.
5. Validate the session before confirming the balance online.

The player's legitimate offline progress should remain available, but suspicious sessions can be excluded from competitive or server-backed currency.

Infinity Taps itself is noncompetitive, but its rewards affect the shared cosmetic economy.

---

### # 8. World Structure

The world should feel extremely large without requiring one enormous hand-built map.

Use a combination of:

- Authored biome scenes
- Procedural background segments
- Landmark scenes
- Transition scenes
- Rare-event scenes
- Reusable environmental decoration
- Distance and altitude coordinates

The player's location can be represented by:

```
```text
world_id
biome_id
horizontal_distance
altitude
depth
local_segment
discovered_landmarks
```
```

### ## Starting Area

The player begins in a peaceful field containing:

- Grass
- Flowers
- Small hills
- Clouds
- Butterflies
- Occasional animals
- A simple ground surface
- Clear directional movement

This area teaches movement without a formal tutorial.

---

## # 9. Suggested Biomes and Regions

### ## Surface Regions

#### ### Basic Field

The default starting environment.

Possible encounters:

- Rabbits
- Butterflies
- Birds
- Traveling slime characters

- Picnic objects
- Rare rainbow
- Small abandoned treasure box containing profile collectibles

#### #### Farmland

Possible elements:

- Crops
- Barns
- Farm animals
- Tractors
- Scarecrows
- Fruit-themed characters
- Seasonal harvest events

#### #### Forest

Possible elements:

- Dense trees
- Mushrooms
- Fireflies
- Owls
- Hidden paths
- Ancient tree landmarks
- Rare forest spirits

#### #### Jungle

Possible elements:

- Vines
- Tropical plants
- Parrots
- Monkeys
- Waterfalls
- Temple ruins
- Rare glowing insects

#### #### Mountain Region

Possible elements:

- Steep slopes
- Snowy peaks
- Eagles
- Mountain goats
- Caves
- High-altitude clouds
- Rare summit encounters

#### ### Desert

Possible elements:

- Sand dunes
- Cacti
- Desert ruins
- Camels
- Dust storms
- Mirages
- Rare oasis scenes

#### ### Volcanic Region

Possible elements:

- Lava
- Ash
- Black rock
- Fire creatures
- Volcanic eruptions
- Obsidian caves
- Rare phoenix encounter

#### ### Arctic and Antarctic Regions

Possible elements:

- Snow
- Icebergs
- Penguins
- Seals
- Polar wildlife appropriate to the region
- Aurora effects
- Ice caves
- Rare frozen characters

The Arctic and Antarctic should be treated as separate regions rather than one combined biome.

---

## # 10. Ocean Exploration

Players should be able to reach an ocean, enter the water, and descend through progressively deeper zones.

### ## Ocean Layers

#### ### Surface Water

- Waves
- Boats
- Seagulls
- Dolphins
- Floating objects
- Islands

#### ### Sunlight Zone

- Colorful fish
- Coral
- Sea turtles
- Rays
- Schools of fish

#### ### Twilight Zone

- Dimmer lighting
- Larger fish
- Jellyfish
- Bioluminescent life
- Unusual sounds

#### ### Midnight Zone

- Near-total darkness
- Anglerfish
- Strange deep-sea creatures
- Sparse light sources

### ### Abyssal Zone

- Slow movement
- Rare wildlife
- Underwater vents
- Alien-looking organisms

### ### Trench Region

The deepest destination can be inspired by the Mariana Trench.

Possible discoveries:

- Unique deep-sea characters
- Ancient structures
- Rare tap effects
- Giant creature silhouettes
- Secret profile badges
- A depth milestone

The player should never run out of air. Breathing and water pressure do not create failure conditions.

---

## # 11. Sky and Space Exploration

Repeated upward movement should allow the player to leave the ground biome and pass through several altitude layers.

### ## Altitude Progression

1. Tree level
2. Cloud layer
3. Storm-cloud layer
4. Upper atmosphere
5. Near-space
6. Orbit
7. Open space
8. Moon
9. Nearby planets
10. Deep-space regions



## ## Sky Encounters

- Birds
- Hot-air balloons
- Aircraft
- Storm clouds
- Rainbows
- Floating islands
- Sky characters
- Rare shooting stars

## ## Space Encounters

- Satellites
- Space stations
- Meteors
- Comets
- Asteroids
- Astronaut characters
- Alien ships
- Unexplained signals
- Rare cosmic creatures

Alien appearances should remain uncommon. Their rarity makes them memorable rather than routine.

Examples:

- An alien ship briefly crosses the background.
- An alien waves through a window.
- A tractor beam lifts Velvet Hex.
- A rare alien NPC offers a collectible profile entry.
- A mysterious creature appears only once per account unless revisited through a collection menu.

---

## # 12. Other Worlds and Planets

Planets should not simply be background objects. Some can be entered and explored.

Potential worlds:

- Moon

- Mars-inspired desert planet
- Ice planet
- Slime planet
- Crystal planet
- Gas-giant floating platforms
- Artificial machine world
- Low-gravity garden world
- Dark void world
- Candy-colored alien world

Each world should have:

- A distinct visual palette
- Different ambient audio
- Unique background objects
- At least one unique character
- A small set of rare discoveries
- One or more landmarks
- A dedicated collection entry

The system can begin with Earth, the ocean, the Moon, and one alien planet. Additional worlds can be added in updates.

---

### # 13. Characters and Encounters

Players may encounter other unique characters while exploring.

These characters can:

- Wave
- Bounce beside the player
- Follow temporarily
- Perform an animation
- Offer a short line of dialogue
- Appear only under certain conditions
- Add an entry to the discovery collection
- Reveal a secret route
- Trigger an environmental event
- Occasionally provide a cosmetic discovery

Most encounters should not stop play. Dialogue should appear in small bubbles and disappear automatically.

## ## Encounter Types

- Common ambient character
- Uncommon named character
- Rare traveling character
- Biome-exclusive character
- Time-limited seasonal character
- Secret character
- Extremely rare legendary encounter

Example rarity structure:

| Encounter tier | Suggested frequency                       |
|----------------|-------------------------------------------|
| Common         | Every few minutes                         |
| Uncommon       | Several times per session                 |
| Rare           | Approximately once across longer sessions |
| Legendary      | Extremely uncommon and memorable          |

Exact rates should be tuned through playtesting rather than fixed permanently.

---

## # 14. Rare Cosmetic Discoveries

Rare tap effects, particle effects, profile items, and other cosmetics can be found through exploration.

A raw **\*\*0.001% chance per tap\*\*** equals approximately one success per 100,000 taps on average. This is rare enough to feel special, but pure random chance could allow one player to receive it immediately while another receives nothing after hundreds of thousands of taps.

A better implementation is an encounter-based system.

## ## Recommended Rare-Spawn Structure

The game should periodically perform an eligible discovery roll based on:

- Valid taps
- Distance traveled
- Current biome
- Whether the player is actively moving
- Existing ownership

- Time since last rare discovery
- Current pity progress

Example:

- Every 100 valid taps, perform one rare-discovery check.
- Base chance per check: 0.1%.
- This still averages roughly one discovery per 100,000 taps.
- Gradually increase the chance after long periods without success.
- Reset the pity counter after a successful discovery.

This prevents constant random-number checks and makes balancing easier.

### ## Discovery Requirements

Some rewards may require:

- Reaching a specific world
- Traveling at a particular depth
- Tapping during a rare weather event
- Encountering a specific character
- Visiting at night
- Accumulating a certain number of taps
- Wearing a specific character skin
- Discovering multiple connected landmarks

### ## Collecting the Reward

A rare cosmetic should appear physically in the environment. The player must tap or move into it to collect it.

The item should:

- Remain available long enough to notice
- Make a distinctive sound
- Glow or animate
- Pause despawning while the game is backgrounded
- Enter the player's permanent collection once touched

Do not allow a rare cosmetic to disappear instantly because the player tapped in the wrong direction.

---

## # 15. Discovery Collection

Infinity Taps should have a **Discovery Journal** containing:

- Biomes visited
- Worlds visited
- Named characters encountered
- Animals observed
- Landmarks discovered
- Rare events witnessed
- Cosmetics found
- Highest altitude
- Greatest ocean depth
- Total distance traveled
- Tap milestones

Locked discoveries can appear as silhouettes or question marks.

Example:

- > **Unknown Deep-Sea Creature**
- > Found somewhere below the Midnight Zone.

The journal gives exploration value even when no Hero Coins are spawning.

---

## # 16. Infinity Taps Achievements

Potential achievements include:

| Achievement              | Requirement                    | Reward                       |
|--------------------------|--------------------------------|------------------------------|
| ---                      | ---                            | ---                          |
| <b>Just One More Tap</b> | Reach 1,000 Infinity Taps.     | Fidget Badge                 |
| <b>Tap Enthusiast</b>    | Reach 10,000 Infinity Taps.    | 10 Hero Coins                |
| <b>Endless Motion</b>    | Reach 100,000 Infinity Taps.   | Endless Tap Badge            |
| <b>Million-Tap Hero</b>  | Reach 1,000,000 Infinity Taps. | Exclusive Tap Effect + Badge |
| <b>World Traveler</b>    | Discover ten biomes.           | Explorer Badge               |
| <b>Into the Clouds</b>   | Reach the cloud layer.         | Cloud Badge                  |
| <b>Beyond the Sky</b>    | Reach space.                   | Astronaut Badge              |
| <b>One Small Tap</b>     | Land on the Moon.              | Moon Badge                   |
| <b>Going Deep</b>        | Reach the Midnight Zone.       | Deep Ocean Badge             |
| <b>At the Bottom</b>     | Reach the deepest trench.      | Abyss Badge                  |
| <b>First Contact</b>     | Encounter an alien.            | Alien Badge                  |

| **\*\*What Are the Odds?\*\*** | Find an extremely rare cosmetic. | Lucky Tapper Badge |  
| **\*\*New Friend\*\*** | Encounter a named character. | Friendship Badge |  
| **\*\*Around the World\*\*** | Visit every initial Earth biome. | Globe Badge |  
| **\*\*Fidget Session\*\*** | Tap 5,000 times during one session. | Fidget Master Badge |

Coin rewards for tap milestones must not double-count the normal one-coin-per-1,000-taps payout unless the achievement explicitly provides an additional bonus.

---

## # 17. Cosmetic Integration

Infinity Taps should showcase all cosmetic purchases and unlocks.

Supported cosmetic slots:

- Character
- Character skin
- Particle trail
- Tap effect
- Movement effect
- Water movement effect
- Landing effect
- Profile badge
- Background companion
- Emote

Tap effects are especially valuable in this mode because the player may see them thousands of times.

The player should be able to adjust effect intensity:

```
```text  
Full | Reduced | Minimal | Off  
```
```

Reduced and minimal modes are important for:

- Photosensitivity
- Visual clarity
- Battery use
- Older mobile devices
- Long fidget sessions

---

## # 18. Audio and Haptic Design

Infinity Taps should feel satisfying even when no rare item appears.

Each tap may produce:

- A soft sound
- A subtle haptic response
- A character wobble
- A small particle
- Environmental movement
- A changing musical layer

Players must be able to independently control:

- Music
- Ambient sounds
- Tap sounds
- Character sounds
- Haptics
- Cosmetic effect sounds

Repeated tap sounds should include slight pitch and sample variation so they do not become irritating.

The game could offer optional fidget sound themes:

- Soft slime
- Retro arcade
- Bubble pop
- Wooden click
- Gentle chime
- Rain drop

These themes can be unlockable cosmetics, provided they offer no gameplay advantage.

---

## # 19. World-Generation Architecture

The world should use loaded segments rather than keeping every biome active simultaneously.

Recommended structure:

```
```text
InfinityTapsManager
WorldTravelManager
BiomeManager
EncounterManager
DiscoveryManager
TapCounterManager
RewardManager
SaveService
```
```

## ## Suggested Godot Scene Structure

```
```text
InfinityTaps.tscn
├── WorldRoot
│   ├── ActiveBiome
│   ├── PreviousSegment
│   └── NextSegment
├── Player
├── Camera2D
├── EncounterLayer
├── EffectsLayer
├── UILayer
│   ├── TapCounter
│   ├── LocationDisplay
│   ├── DiscoveryPopup
│   └── CoinProgress
└── InfinityTapsManager
```
```

The game should preload the next segment before the player reaches the current segment boundary.

---

## # 20. Tap Event Structure

Every valid tap should generate a central event.

```
```gdscript
signal infinity_tap_registered(
```



```
    screen_position: Vector2,  
    world_position: Vector2,  
    direction: Vector2,  
    total_taps: int  
)  
...
```

Processing order:

1. Validate tap.
2. Increment session taps.
3. Increment lifetime taps.
4. Calculate direction.
5. Move or deform the character.
6. Trigger equipped tap effect.
7. Update coin progress.
8. Update achievement progress.
9. Update distance and location.
10. Evaluate encounters and discoveries.
11. Schedule save operation.

Do not write the save file after every individual tap. Save using:

- Timed intervals
- Milestone completion
- Biome transition
- Rare discovery
- Application pause
- Mode exit

Keep a small in-memory journal so recent taps can be restored if the application closes unexpectedly.

21. Save Data

Store at minimum:

```
```text  
infinity_lifetime_taps
infinity_session_best
infinity_coin_milestones_paid
infinity_tap_remainder
```

```
current_world_id
current_biome_id
horizontal_distance
current_altitude
current_depth
highest_altitude
greatest_depth
discovered_biomes
discovered_worlds
discovered_characters
discovered_landmarks
discovered_events
found_cosmetic_ids
rare_discovery_pity
last_save_timestamp

```

The paid tap milestone should be derived safely:

```
```text
coins_earned_from_taps = floor(valid_lifetime_taps / 1000)
---
```

The game must compare this against previously granted milestone rewards before issuing additional coins.

22. Recommended Development Phases

Phase 1 — Fidgeting MVP

Build:

- Persistent tap counter
- One coin per 1,000 valid taps
- Left, right, and upward movement
- No-loss gameplay
- Basic field biome
- Clouds and simple sky
- Cosmetic tap effects
- Local saving
- Tap milestone display
- Basic ambient encounters

The purpose of Phase 1 is to determine whether tapping and movement feel satisfying without relying on rewards.

Phase 2 — Earth Exploration

Add:

- Forest
- Farmland
- Mountain
- Desert
- Ocean surface
- Underwater movement
- Initial discovery journal
- Named characters
- Landmarks
- Biome transitions

Phase 3 — Vertical Exploration

Add:

- Upper atmosphere
- Space
- Moon
- Ocean depth layers
- Deep trench
- Altitude and depth milestones
- Rare environmental events

Phase 4 — Rare Discoveries

Add:

- Cosmetic world spawns
- Pity system
- Rare character encounters
- Secret conditions
- Collection completion rewards
- Seasonal discoveries

Phase 5 — Expanded Universe

Add:

- Additional planets
- Alien encounters
- Exotic worlds
- World-specific creatures
- More dynamic events
- New exploration routes
- Larger discovery journal

23. Important Design Boundaries

Infinity Taps should not become another survival mode.

Avoid adding:

- Increasing gravity
- Lives
- Health
- Timers
- Forced objectives
- Competitive rankings
- Punishing hazards
- Energy systems
- Paywalled travel
- Required advertisements
- Coin multipliers sold for money

The mode succeeds when players feel comfortable opening it without committing to a formal game session.

Final Mode Identity

Infinity Taps should feel like a combination of:

- A digital fidget toy
- An endless exploration game
- A cosmetic showcase
- A discovery collection
- A quiet companion experience

The player is never told that they failed. Every tap adds to permanent progress, every direction may reveal something new, and rare discoveries make long-term casual play feel meaningful.

Restricted rewards

Infinity Taps — Exclusive Earned Cosmetics

Infinity Taps cosmetics must be treated as **gameplay-earned exclusives**. They cannot be purchased with Hero Coins, real money, seasonal passes, supporter bundles, or direct store offers.

This gives Infinity Taps its own progression identity and ensures players who prefer relaxed exploration still receive meaningful rewards that reflect how they choose to play.

Design Purpose

These unlocks serve several goals:

- Reward players who primarily enjoy noncompetitive play
- Give Infinity Taps long-term progression beyond the tap counter
- Create visible proof that a player explored rare locations or invested time in the mode
- Make discoveries feel meaningful because they cannot simply be bought
- Give collectors reasons to engage with multiple game modes
- Let casual players earn cosmetics that competitive players may not have

A player wearing an Infinity Taps skin or effect should communicate:

“I found or earned this through exploration.”

The value comes from the experience attached to the item, not from its monetary price.

Exclusivity Rules

Infinity Taps rewards must never be available through:

- The standard cosmetic shop
- Hero Coin purchases
- Season passes
- Supporter bundles
- Paid character bundles
- Rewarded advertisements
- Promotional purchase packages
- Random paid loot systems

They may only be unlocked through Infinity Taps activities such as:

- Tap milestones
- Biome discoveries
- Rare world spawns
- Character encounters
- Exploration achievements
- Hidden environmental conditions
- Altitude or depth milestones
- Discovery Journal completion
- Mode-specific events

The system should clearly label them:

Infinity Taps Exclusive

Earned through exploration. Not available for purchase.

Reward Categories

Infinity Taps may award:

- Character skins
- Tap effects
- Particle trails
- Landing effects
- Movement effects
- Water trails
- Space trails
- Profile badges
- Profile frames
- Character colors
- Companion creatures
- Ambient sound themes
- Emotes
- Character idle animations

The strongest rewards should be tied to meaningful discoveries rather than only raw tapping totals.

Example Infinity Taps Exclusives

Reward	Unlock Requirement
Meadow Velvet Skin	Discover every landmark in the starting field.
Forest Spirit Trail	Encounter the hidden forest spirit.
Volcanic Tap Effect	Reach the deepest volcanic chamber.
Abyssal Velvet Skin	Reach the deepest ocean trench.
Bioluminescent Trail	Discover five deep-sea creatures.
Moon Dust Tap Effect	Land on the Moon.
Alien Signal Effect	Witness a rare alien transmission.
First Contact Skin	Encounter an alien character.
World Traveler Frame	Visit every launch-version biome.
Endless Ripple Effect	Reach 100,000 Infinity Taps.
Million Tap Aura	Reach 1,000,000 Infinity Taps.
Cosmic Velvet Skin	Discover all initial space regions.

Rare Discovery Rewards

Some effects and skins may spawn in the world at very low rates.

When one appears:

- It must be visually distinctive.
- It should remain available long enough to collect.
- The game should save the spawn immediately.
- Closing the application should not permanently destroy the opportunity.
- Collecting it should unlock the cosmetic permanently.
- Duplicate copies should not repeatedly appear unless they provide another reward.

For exceptionally rare rewards, a hidden pity system should prevent impossible bad luck.

For example:

Base discovery chance: 0.001% per eligible tap
 Chance increases slowly after 50,000 eligible taps
 Guaranteed discovery by a configured maximum threshold

The exact pity threshold should remain hidden, but the player should never be able to tap indefinitely with no realistic possibility of receiving the reward.

Different Unlock Types

Infinity Taps rewards should use several acquisition methods so the mode does not become only a tap-count grind.

Tap Milestones

Awarded for cumulative taps:

- 1,000 taps
- 10,000 taps
- 100,000 taps
- 500,000 taps
- 1,000,000 taps

Exploration Milestones

Awarded for reaching places:

- Cloud layer
- Upper atmosphere
- Space
- Moon
- Ocean Midnight Zone
- Deepest trench
- Volcano core
- Ice cave
- Alien planet

Discovery Rewards

Awarded for finding:

- Rare animals
- Named characters
- Hidden landmarks
- Environmental events
- Secret routes

- Unusual weather
- Cosmic encounters

Collection Rewards

Awarded for completing sets:

- Every creature in one biome
- Every Earth biome
- Every ocean depth
- Every launch-version planet
- Every rare field event

Conditional Secrets

Awarded through unusual behavior:

- Tap a meteor as it crosses the screen.
- Follow a whale through several ocean segments.
- Remain beside a sleeping character without leaving.
- Reach space using only upward taps.
- Travel from the highest point to the deepest trench.
- Encounter the same traveler in multiple worlds.

Profile and Multiplayer Visibility

Infinity Taps cosmetics should remain usable throughout the entire game.

A player may equip them in:

- Standard local survival
- Online multiplayer
- Profiles
- Leaderboards
- Match introductions
- Victory screens
- Infinity Taps itself

This gives noncompetitive players visible recognition outside their preferred mode.

For example, a player using the Abyssal Velvet Skin in multiplayer signals that they reached the deepest ocean region, even if they rarely play competitive matches.

The cosmetic should show its source when inspected:

Abyssal Velvet

Reached the deepest trench in Infinity Taps.

Infinity Taps Exclusive — Cannot be purchased.

Shop Behavior

Infinity Taps exclusives may appear in the cosmetic collection interface, but they must not display a price or purchase button.

Instead, show:

- Unlock requirement
- Discovery hint
- Current progress
- Source mode
- Locked silhouette where appropriate

Example:

Moon Dust Tap Effect

Locked

Hint: Travel beyond Earth and touch another world.

Extremely secret rewards may hide their exact unlock conditions until discovered.

The shop may display a preview, but it must direct players to Infinity Taps rather than offering a purchase.

Available through Infinity Taps exploration

This can encourage players to try the mode without undermining the exclusive nature of the reward.

Data Requirements

Every cosmetic definition should include an acquisition source.

```
{  
  "cosmetic_id": "skin_velvet_abyssal",
```

```
"display_name": "Abyssal Velvet",
"cosmetic_type": "character_skin",
"acquisition_type": "infinity_taps",
"purchasable": false,
"tradable": false,
"unlock_condition_id": "reach_deepest_trench",
"exclusive_mode": "infinity_taps",
"permanent_unlock": true
}
```

Important fields:

```
acquisition_type
purchasable
exclusive_mode
unlock_condition_id
discovery_rarity
hidden_until_discovered
permanent_unlock
seasonal
```

The store service must reject any attempt to assign a price to an item where:

```
purchasable == false
```

Reward Granting

When the player meets the requirement:

1. Verify that the unlock condition was completed.
2. Confirm that the cosmetic is not already owned.
3. Add it to the permanent cosmetic inventory.
4. Save immediately.
5. Record the discovery location and timestamp.
6. Trigger the unlock presentation.
7. Add the item to the Discovery Journal.
8. Make it available in all relevant loadout menus.

The unlock should use a unique transaction key:

```
infinity_unlock:skin_velvet_abyssal
```

This prevents duplicate grants during reconnects or save retries.

Offline and Online Validation

Infinity Taps should remain playable offline. Locally earned cosmetics can be stored as pending unlocks and synchronized later.

For common milestone items, local validation is sufficient during early development.

For very rare or prestigious rewards, the online version should validate:

- Tap totals
- Discovery conditions
- World coordinates
- Encounter IDs
- Session duration
- Impossible tap rates
- Duplicate reward transactions

The goal is not to make Infinity Taps feel restrictive. Validation protects the meaning of rare items once they become visible on public profiles and in multiplayer.

Avoiding Unhealthy Scarcity

Infinity Taps exclusives should generally remain permanently obtainable. Their rarity should come from exploration, effort, or discovery—not from requiring players to participate during a narrow purchase window.

Seasonal Infinity Taps events can exist, but permanent rewards should form the foundation of the mode.

This creates a useful distinction:

- **Store cosmetics:** purchased
- **Seasonal cosmetics:** earned during a limited season
- **Competitive cosmetics:** earned through survival or multiplayer
- **Infinity Taps cosmetics:** earned through tapping and exploration

Each area of the game gives players its own recognizable reward path.

Final Design Principle

Infinity Taps is not a secondary mode that only pays small amounts of currency. It has its own exclusive collection and status system.

Players who choose calm, noncompetitive exploration should be able to earn some of the most interesting cosmetics in the game. Those rewards become valuable because they represent time spent exploring strange places, discovering rare events, and participating in a mode that other players may overlook.

Size and boosts

Two-minute zones are a strong target for Infinity Taps. That is long enough for players to hear most of a biome's musical loop, interact with the environment, and discover something, without making movement between zones feel slow.

Each zone could follow a simple rhythm:

- **0:00–0:30:** establish the biome, music, wildlife, and environmental interactions
- **0:30–1:30:** exploration, collectibles, heroes, secrets, and temporary boosts
- **1:30–2:00:** visual or musical transition toward the next zone

The boosts fit Infinity Taps well because they change how tapping feels without introducing failure.

Infinity Boosts

- **Velocity Increase:** temporarily increases horizontal movement and momentum, letting the player race through part of the biome.
- **Jet Engine:** holding the screen activates continuous upward thrust. Fuel or heat could limit how long it lasts.
- **Anti-Gravity:** reverses normal controls so every tap pushes the player downward instead of upward.
- **Extra Bouncy:** dramatically increases jump height and ground bounce, launching the character absurdly high.
- **Heavy Gravity:** taps produce smaller jumps, but landing creates stronger impacts and environmental reactions.
- **Tap Burst:** every tap releases a short shockwave that interacts with nearby leaves, water, bushes, and animals.
- **Moon Bounce:** lowers gravity and creates slow, exaggerated aerial movement.
- **Magnetism:** pulls nearby Infinity-specific collectibles or hidden objects toward the player.

Boosts should last approximately **10–25 seconds**, with the music temporarily adding a matching layer. The jet engine could introduce a rising synth, anti-gravity could reverse part of the melody, and extra bouncy could add spring-like percussion. This would make each boost feel like a brief playable remix of the biome.

Monetization Method

Tap Against Me — Updated Monetization Model

This creates a fair **cosmetic-first monetization system** without selling power, higher jump force, lower gravity, better fruit odds, or competitive advantages.

Core Monetization Rule

Paid purchases may change how the player, profile, taps, particles, or effects look, but must never improve gameplay performance.

Paid content must not provide:

- Increased jump force
- Reduced gravity
- Extra health or revives in competitive modes
- Better item spawn rates
- Faster achievement progression
- Ranked advantages
- Exclusive gameplay abilities
- Stronger characters

All characters and skins should use the same underlying gameplay rules unless character differences are available equally through normal gameplay and carefully balanced.

Initial Store Products

Product	Price	Contents
Hero Coin Starter Pack	\$0.99	Small Hero Coin package intended to help new players purchase their first cosmetics
30-Day Ad-Free Pass	\$2.99	Removes non-optional advertisements for 30 days
Season Pass	\$4.99	Unlockable seasonal cosmetics available only during that season

Supporter Bundle	\$9.99	Character bundle, character particle effect, tap effect, and 60 days of ad-free access
Character Skins	Variable	Cosmetic visual changes for existing characters
Particle Effects	Variable	Trails, movement particles, landing effects, and elimination effects
Tap Effects	Variable	Visual and audio feedback triggered when the player taps

30-Day Ad-Free Pass

The **30-Day Ad-Free Pass** should cost **\$2.99** and begin immediately when purchased.

It removes:

- Interstitial advertisements
- Advertisements shown between rounds
- Advertisements shown when navigating menus
- Other mandatory advertising placements

It should not disable optional rewarded advertisements. Players with an active ad-free pass should still be able to voluntarily watch a rewarded ad when they want the associated reward.

Examples:

- Double post-round Hero Coins
- One solo-mode revival
- Temporary cosmetic trial
- Daily challenge reroll

The game should clearly display the expiration date:

Ad-Free Until: September 14, 2026

Players should receive reminders:

- Seven days before expiration
- One day before expiration
- After the pass expires

The pass should not automatically renew unless the player explicitly purchases a subscription product through the platform. The initial implementation can simply be a manually repurchased 30-day entitlement.

Hero Coin Starter Pack

The **\$0.99 Hero Coin Starter Pack** gives new players enough currency to interact with the cosmetic shop early.

The goal is not to sell a large currency balance. It should allow the player to purchase approximately:

- One basic skin, or
- One tap effect, or
- One small combination of inexpensive cosmetics

The exact amount depends on the final shop economy, but an example would be:

Starter Pack — \$0.99

150 Hero Coins

One purchase per account

Making this a one-time starter offer increases its perceived value without undermining larger cosmetic purchases.

A repeatable standard Hero Coin package can exist separately later, but the starter package should remain visibly better than ordinary currency packs.

Seasonal Pass

Each season lasts approximately three months and has a **\$4.99 seasonal pass**.

The pass should contain only cosmetic and profile rewards, including:

- Seasonal character skins
- Character color variants
- Tap effects
- Particle trails
- Landing effects

- Profile frames
- Profile badges
- Emotes
- Victory animations
- Hero Coins
- Seasonal title cards

Pass Structure

The season should have two reward tracks:

Free Track

Available to every player.

Possible rewards:

- Hero Coins
- One basic seasonal cosmetic
- Standard badges
- Challenge rewards
- Small cosmetic items

Premium Track

Unlocked for \$4.99.

Possible rewards:

- Premium character skin
- Exclusive particle effect
- Exclusive tap effect
- Animated profile badge
- Premium profile frame
- Additional Hero Coins
- Final season completion cosmetic

Players earn seasonal progress through gameplay and challenges. Purchasing the pass should not instantly grant every item. The player unlocks rewards by playing during the season.

Seasonal Exclusivity

Seasonal cosmetics are obtainable only during their original season.

Once the season ends:

- Players retain everything they unlocked.
- Unfinished rewards become unavailable.
- Seasonal rewards do not enter the normal shop.
- Cosmetics do not provide gameplay advantages.
- Profiles should show the season associated with each item.

This makes the cosmetics meaningful collectibles while keeping competition fair.

The game should clearly communicate the remaining season duration and required progress. Players should never be surprised by a pass expiring.

Supporter Bundle

The **\$9.99 Supporter Bundle** is intended for players who want to support development and receive a coordinated cosmetic package.

Recommended Contents

- One exclusive character or character skin bundle
- One matching character particle effect
- One matching tap effect
- 60 days of ad-free access
- Supporter profile badge
- Optional small Hero Coin bonus

Example:

Velvet Hex Supporter Bundle — \$9.99

- Supporter Velvet Hex Skin
- Shadow Slime Particle Trail
- Purple Hex Tap Effect
- Supporter Badge
- 60 Days Ad-Free
- 100 Hero Coins

The bundle should not include stronger character abilities, bonus gravity resistance, additional competitive lives, or superior fruit effects.

I would make the founding Supporter Bundle a **one-time account purchase**. Additional themed supporter bundles could be offered later without reusing the same exclusive badge.

Cosmetic Categories

Character Skins

Character skins change a character's appearance while preserving the same gameplay behavior.

Examples for Velvet Hex:

- Toxic Velvet
- Golden Velvet
- Arcade Velvet
- Frozen Velvet
- Molten Velvet
- Ghost Velvet
- Corrupted Velvet
- Strawberry Slime Velvet

Skins may alter:

- Body colors
- Facial details
- Horn appearance
- Tail shape
- Idle animation
- Jump animation appearance
- Cosmetic landing animation

They should not alter the collision shape or make the character visually misleading in competitive gameplay.

Character Particle Effects

These remain attached to or trail behind the character.

Examples:

- Purple smoke

- Golden sparkles
- Slime droplets
- Pixel stars
- Poison bubbles
- Tiny ghosts
- Lightning pixels
- Falling flower petals

Effects should have a reduced-intensity option for accessibility and performance.

Tap Effects

Tap effects appear whenever the player taps.

Examples:

- Pixel shockwave
- Purple claw mark
- Slime splash
- Golden ring
- Tiny lightning strike
- Heart burst
- Retro arcade star
- Poison ripple

Tap effects may include sound, but players should be able to independently reduce or disable cosmetic sound effects.

Other Cosmetic Opportunities

The store can later support:

- Background themes
- Arena themes
- Fruit skins
- Ground impact effects
- Elimination effects
- Username colors
- Profile frames
- Player titles
- Emotes
- Victory poses
- Match entrance animations

Advertising Rules

Advertisements should never interrupt a live round.

Recommended placements:

- After several completed solo rounds
- After returning to the main menu
- Optional post-round reward multiplier
- Optional solo revival
- Optional cosmetic preview reward

Avoid:

- Ads after every loss
- Ads during online matchmaking
- Ads before a player can claim an earned achievement
- Ads that disguise themselves as interface buttons
- Forced ads immediately after a purchase
- Ads for players with an active ad-free entitlement

A reasonable starting rule would be no more than one forced advertisement after every three to five solo rounds, with a cooldown measured in real time.

Online competitive play should ideally remain free from forced ads between consecutive matches so players can rematch quickly.

Store Presentation

The shop should organize products into clear categories:

Featured | Characters | Skins | Tap Effects | Particles | Profile | Hero Coins

Each cosmetic should show:

- Preview
- Character compatibility
- Price

- Ownership status
- Rarity
- Source
- Seasonal status
- Equip button
- Purchase confirmation

Players should be able to preview animations and effects before purchasing.

For tap effects, the preview should allow the player to tap repeatedly inside a demonstration area.

For particle effects, the preview should show the character moving and jumping.

Hero Coin Economy

Hero Coins may be earned through:

- Gameplay pickups
- Achievements
- Daily challenges
- Weekly challenges
- Seasonal rewards
- Optional rewarded ads
- Paid currency packages

The free earning rate should be slow enough that cosmetics remain meaningful, but not so slow that the shop feels inaccessible without payment.

A healthy early economy would allow an active free player to earn:

- A small cosmetic every few days
- A larger character skin after one to three weeks
- Occasional premium-looking rewards through achievements and seasonal play

Paid purchases should accelerate cosmetic collection, not determine whether the player can enjoy the game.

Purchase and Ownership Data

Each product should have a permanent internal ID.

Example:

hero_coins_starter_001
ad_free_30_days
season_pass_2026_03
supporter_bundle_founder
skin_velvet_toxic
particle_velvet_shadow
tap_effect_hex_burst

The player profile should store:

owned_cosmetic_ids
equipped_skin_id
equipped_particle_id
equipped_tap_effect_id
hero_coin_balance
active_season_pass_id
ad_free_expires_at
completed_purchase_ids
supporter_status

The ad-free expiration must use a trusted server or platform timestamp where possible. It should not rely solely on the phone's local clock.

Developer Safeguards

The purchase system must:

- Validate purchases through Google Play or Apple
- Restore purchases after reinstalling
- Prevent duplicate rewards
- Track consumable and non-consumable products differently
- Store seasonal entitlement separately from progress
- Grant 30 or 60 days from the purchase timestamp
- Preserve cosmetics permanently
- Never remove already unlocked seasonal items
- Handle purchase cancellation or failed validation

- Log every Hero Coin currency transaction
- Keep paid gameplay advantages out of all store definitions

The Supporter Bundle's cosmetics are permanent, while its ad-free component expires after 60 days.

The Season Pass expires at the season's end, but all cosmetics already earned from it remain owned permanently.

Final Monetization Position

The game's value proposition becomes:

Play for free, earn cosmetics through skill and achievements, or spend money to personalize characters and support continued development.

This model gives players several reasons to purchase without undermining competition:

- The \$0.99 starter pack creates an easy first purchase.
- The \$2.99 ad-free pass improves the experience without becoming a permanent entitlement.
- The \$4.99 seasonal pass supports recurring engagement.
- The \$9.99 Supporter Bundle offers clear value to invested players.
- Skins, particles, and tap effects support long-term cosmetic collection.
- No purchase makes a player stronger than someone playing for free.

Recommended Hero Coin Pricing

I would anchor the economy at approximately **100 Hero Coins for \$0.99**. That creates a simple mental conversion without making earned coins feel worthless.

This recommendation assumes an active free player earns roughly **50–100 Hero Coins per day** through gameplay, achievements, and challenges.

Package	Price	Hero Coins	Approx. bonus	Cost per 100
Hero Coin Pocket	\$0.99	100	Base rate	\$0.99
Hero Coin Pouch	\$4.99	600	20%	\$0.83

Hero Coin Chest	\$9.99	1,300	30%	\$0.77
Hero Coin Vault	\$24.99	3,500	40%	\$0.71
Hero Treasury	\$99.99	15,000	50%	\$0.67

The bonus percentages are approximate because the coin totals are rounded into cleaner numbers.

One-Time Starter Offer

Keep the previously discussed starter package separate from the repeatable \$0.99 package:

Hero Starter Pack — \$0.99

200 Hero Coins

One purchase per account

This gives new players twice the normal value and enough currency to purchase an early cosmetic or combine it with earned coins for something better.

After purchasing it, the standard **100 Coins for \$0.99** package replaces it.

Recommended Cosmetic Prices

The coin packs only feel fair when connected to understandable cosmetic prices.

Cosmetic type	Suggested price
Basic color variation	50–100 Coins
Basic tap effect	100–200 Coins
Premium tap effect	250–400 Coins
Basic character particle effect	200–350 Coins
Premium particle effect	400–600 Coins
Standard character skin	400–700 Coins
Premium animated skin	800–1,200 Coins
Character cosmetic bundle	1,200–2,000 Coins

Premium themed collection	2,000–3,500 Coins
---------------------------	----------------------

This would mean:

- **\$0.99** buys a basic cosmetic.
- **\$4.99** buys a standard skin or several smaller effects.
- **\$9.99** buys a premium skin and another effect.
- **\$24.99** buys a substantial themed collection.
- **\$99.99** is for committed collectors purchasing cosmetics over several seasons.

How Free Earnings Should Compare

A player should never feel that earning coins through gameplay is pointless. I would target approximately:

Player activity	Approximate earnings
One casual session	10–25 Coins
Active day	50–100 Coins
Completed daily challenges	15–30 Coins
Completed weekly challenges	75–150 Coins
Average achievement	5–25 Coins
Major achievement	25–100 Coins

At that rate:

- A basic effect takes approximately **one to four active days**.
- A standard skin takes approximately **one to two weeks**.
- A premium animated skin takes approximately **two to four weeks**.
- A player can still earn desirable cosmetics without paying.
- A purchase serves as a shortcut and additional support rather than a requirement.

Important Economy Rule

Paid coins should buy cosmetics, but most cosmetics should also be obtainable with coins earned through gameplay.

There can still be distinct content sources:

- **Normal shop cosmetics:** earned or purchased with Hero Coins.
- **Achievement cosmetics:** earned only by completing achievements.
- **Infinity Taps cosmetics:** earned only by playing Infinity Taps.
- **Seasonal cosmetics:** earned only during their original season.
- **Supporter cosmetics:** included with designated supporter packages.

This prevents direct currency purchases from making every reward meaningless. A player displaying a difficult achievement badge or an extremely rare Infinity Taps effect still demonstrates that they actually earned it.

The \$99.99 Package

I would include the \$99.99 package only after the shop contains enough content to justify it. Fifteen thousand coins should purchase several full character collections, not leave the buyer with nothing meaningful to spend them on.

It should not be labeled **Best Value** or pushed aggressively. Better labels would be:

Hero Treasury

For dedicated collectors and long-term supporters.

The **\$9.99 or \$24.99 package** should be presented as the most popular or recommended option. Those are more realistic purchases for the game's audience.

Seasonal Pass Interaction

The \$4.99 Season Pass should remain a separate direct purchase rather than costing Hero Coins. Otherwise, players could stockpile currency and stop generating recurring seasonal revenue.

The pass can award Hero Coins during progression, but those coins should be a supporting reward rather than enough to purchase the next season automatically.

Final Recommended Structure

- **\$0.99 one-time starter:** 200 Coins
- **\$0.99 repeatable:** 100 Coins
- **\$4.99:** 600 Coins
- **\$9.99:** 1,300 Coins
- **\$24.99:** 3,500 Coins
- **\$99.99:** 15,000 Coins

- **\$2.99:** 30 days without forced ads
- **\$4.99:** Seasonal Pass
- **\$9.99:** Supporter Bundle with cosmetics and 60 days without forced ads

This structure keeps the economy understandable, gives larger purchasers progressively better value, and preserves the importance of gameplay-earned rewards.

For implementation, both Apple and Google provide flexible regional pricing systems, so these should be treated as the U.S. base prices and localized through the storefront rather than manually converted using exchange rates. (developer.apple.com)

Branding

Branding Slogan — Developer Implementation Brief

Official Slogan

Every Tap Against Me Takes You Further.

This is the primary slogan for **Tap Against Me (TAM)**. It represents the complete game rather than only the survival mode:

- Every tap helps the player remain airborne.
- Every tap contributes to progression and achievements.
- Every tap moves the player through Infinity Taps exploration.
- Every tap can lead toward new characters, worlds, cosmetics, and discoveries.

Primary Presentation

Display the slogan with the game title as:

TAP AGAINST ME

Every Tap Against Me Takes You Further.

The slogan should always use the exact wording and capitalization above.

Recommended In-Game Placement

Use the slogan in the following locations:

- Initial loading or splash screen
- Main menu beneath the game logo
- First-launch introduction
- Title screen
- Official website or landing page
- Store listing promotional graphics
- Trailers and social media advertisements
- Major update or seasonal launch screens

The main menu should present it subtly beneath the logo rather than competing with the navigation buttons.

Optional Loading-Screen Treatment

The slogan may appear in stages:

Every Tap
Against Me
Takes You Further.

Alternatively, each section can appear after a light tap animation:

1. **Every Tap**
2. **Against Me**
3. **Takes You Further.**

This should be brief and skippable so it does not delay players entering the game.

Visual Direction

- Use the primary game font or a matching retro pixel font.
- Keep the slogan smaller than the game title.
- Use sufficient contrast for readability.
- Avoid excessive particle effects behind the text.
- A subtle tap ripple, character bounce, or upward movement can accompany its appearance.
- Preserve safe margins for different mobile screen sizes.

Localization

Store the slogan as a localization key rather than placing the text directly inside scenes:

branding.main_slogan

Default English value:

Every Tap Against Me Takes You Further.

Any translated version should preserve the meaning of tapping, progression, and going farther rather than translating each word mechanically.

Short Marketing Variant

For smaller spaces where the full slogan does not fit, use:

Tap. Rise. Explore.

This shortened version should only be used for compact advertising or interface placements.
The official full slogan remains:

Every Tap Against Me Takes You Further.

IMPORTANT

IMPORTANT

Google Project: Joshuaparsonscativity@gmail.com

Online Mode

Tap Against Me

Competitive Ranking, Matchmaking, Bots, and Seasonal Progression Scope

Core Objective

Tap Against Me should provide fast, fair competitive matches regardless of the size of the active player population.

The system should:

- Match players against opponents near their demonstrated skill level.
 - Avoid extreme mismatches such as a 15-second player facing a 72-second player.
 - Use Glicko-2 ratings to track competitive skill.
 - Use survival performance as an additional matchmaking safeguard.
 - Fill empty matchmaking queues with appropriately skilled bots.
 - Provide visible Hearthstone-inspired rank tiers.
 - Reset ranked progression every 90 days.
 - Preserve long-term accomplishments across seasons.
 - Make each opponent feel beatable with slightly improved performance.
-

1. Competitive Rating System

Rating Model

Use **Glicko-2** rather than standard Elo.

Each player should have:

- Rating
- Rating deviation
- Rating volatility
- Ranked match count
- Ranked wins
- Ranked losses
- Current rank tier

- Current rank division
- Highest rank this season
- Highest rank achieved historically
- Current win streak
- Seasonal points or stars
- Placement status

Glicko-2 is preferred because it accounts for rating uncertainty.

A new player, returning player, or player with few matches should have a higher rating deviation. Their rating should move more quickly until the system becomes confident in their skill level.

An active player with many consistent matches should have a lower rating deviation and more stable rating changes.

Initial Rating

Recommended starting values:

Initial rating: 1500

Initial rating deviation: 350

Initial volatility: 0.06

These values may be tuned during testing.

The displayed rank does not need to directly expose the numerical Glicko rating.

2. Visible Rank Structure

The player-facing ranked system should use recognizable competitive tiers.

Recommended rank order:

1. Bronze
2. Silver
3. Gold
4. Platinum
5. Diamond
6. Master
7. Grandmaster

Bronze through Diamond should contain divisions.

Recommended structure:

Bronze V
Bronze IV
Bronze III
Bronze II
Bronze I

Silver V through Silver I
Gold V through Gold I
Platinum V through Platinum I
Diamond V through Diamond I

Master
Grandmaster

Division V is the entry division. Division I is the highest division before promotion into the next tier.

Rank Progression

Use a Hearthstone-inspired star or rank-point system for visible progression.

Example:

Win: Gain 1 rank star
Loss: Lose 1 rank star
Win streak: Gain bonus progression
Reach required stars: Advance one division
Lose all stars at division floor: Risk demotion

Recommended starting requirements:

Rank	Stars per division
Bronze	3
Silver	4
Gold	5

Platinum 5

Diamond 5

Master and Grandmaster should use visible leaderboard placement or rating rather than divisions.

Rank Floors

Players should not be able to fall below certain major milestones during a season.

Recommended rank floors:

- Bronze V
- Silver V
- Gold V
- Platinum V
- Diamond V
- Master

Once a player reaches one of these floors, they cannot fall into the previous tier during that season.

This reduces frustration while still allowing movement within each tier.

3. Relationship Between Glicko and Visible Rank

Glicko-2 should be the primary skill measurement.

The visible rank system should provide understandable seasonal progression.

These systems should be connected but not identical.

Glicko Responsibilities

Glicko determines:

- Opponent strength

- Matchmaking eligibility
- Rating gains and losses
- Leaderboard placement
- Bot skill selection
- Hidden matchmaking rating

Visible Rank Responsibilities

Visible ranks determine:

- Player-facing progression
- Seasonal rewards
- Rank badges
- Profile presentation
- Milestone rewards
- Seasonal goals

A player should generally advance through visible ranks as their Glicko rating improves, but small differences between visible rank and matchmaking rating are acceptable.

This allows the matchmaking system to correct for smurfing, rapidly improving players, returning players, and placement uncertainty.

4. Performance Profile

Glicko alone should not determine matchmaking.

Each player should also have a rolling performance profile.

Track:

- Average survival duration
- Median survival duration
- Average survival duration over the last 10 matches
- Average survival duration over the last 25 matches
- Personal best survival duration
- Recent best survival duration
- Survival-time variance
- Tap frequency
- Tap efficiency
- Recovery success rate

- Item pickup rate
- Item usage effectiveness
- Hazard avoidance
- Disconnect rate
- Abandonment rate

The performance profile is not necessarily visible to the player in full.

Its primary purpose is to stop obviously unfair matches while the Glicko rating is uncertain or temporarily inaccurate.

5. Matchmaking Compatibility

Matchmaking should first search by Glicko rating and then validate the opponent using survival performance.

Recommended compatibility weighting:

70% Glicko rating similarity
20% survival-performance similarity
10% network connection quality

These percentages are initial tuning values.

Survival-Time Compatibility

Example performance groups:

Average survival	Preferred opponent range
0–15 seconds	Within approximately 8 seconds
16–30 seconds	Within approximately 12 seconds
31–50 seconds	Within approximately 15 seconds
51–75 seconds	Within approximately 18 seconds
76+ seconds	Rating and percentile based

These should not be rigid brackets. They should function as matchmaking guidelines.

A player averaging 63 seconds could reasonably face someone averaging 72 seconds.

That matchup gives the lower-performing player a realistic benchmark and an opportunity to improve.

A player averaging 15 seconds should not normally face someone averaging 72 seconds.

That gap is not meaningfully competitive.

Match Quality Principle

A good match should feel like:

I can beat this opponent if I perform slightly better than I normally do.

The system should target an expected win probability near 50%.

An acceptable general range is approximately:

Expected win probability: 40% to 60%

Wider ranges may be permitted after longer queue times.

6. Matchmaking Search Expansion

Matchmaking should gradually broaden the acceptable opponent range.

Recommended flow:

0–5 Seconds

Search for:

- Very close Glicko rating
- Similar survival average
- Good connection quality
- Real players only

5–10 Seconds

Expand:

- Glicko range slightly
- Survival-time range slightly
- Regional search area where latency remains acceptable

10–15 Seconds

Expand further:

- Wider Glicko range
- Wider performance range
- Continue prioritizing real players

After 15 Seconds

Insert an appropriately rated bot unless the player selected a real-player-only option.

Exact timings should be configurable remotely.

Player Setting

Players may have the option:

Allow Bot Opponents: On / Off

Turning bot opponents off may result in longer queue times.

Ranked modes may override this option in lower population situations, depending on the final ranked rules.

7. Bot Matchmaking

Bots should be treated as rated competitive participants.

Each bot profile should have:

- Internal Glicko rating
- Target survival range
- Reaction-time range
- Awareness radius
- Tap consistency
- Error frequency
- Recovery skill
- Item recognition speed
- Item pursuit behavior
- Risk tolerance
- Preferred play style
- Match history
- Performance statistics

Bots should be selected based on the player's hidden rating and recent performance.

Bot Performance Targeting

A bot should not be programmed to fail at an exact predetermined time.

Instead, it should produce a realistic distribution.

Example for a player averaging 63 seconds:

Bot average target: 60–68 seconds

Common result range: 52–72 seconds

Occasional early failure: 40–50 seconds

Occasional strong result: 73–82 seconds

The bot should not always outperform or underperform the player in the same pattern.

8. Bot Behavior and Human Imperfection

Bots must use the same gameplay rules as human players.

They should not have:

- Perfect frame-level reactions
- Hidden movement advantages
- Impossible item awareness

- Zero reaction delay
- Guaranteed recovery
- Knowledge of future item spawns
- Physics unavailable to the player

Difficulty should come from:

- Better timing
- More consistent movement
- Improved recovery
- Better item evaluation
- Stronger risk management
- Fewer mistakes

Human-Like Errors

Bots should occasionally:

- Tap too early
- Tap too late
- Overcorrect
- Under-correct
- Panic near the ground
- Misjudge momentum
- Miss an item
- Chase an item too aggressively
- Ignore a useful item
- Use an item inefficiently
- Fail a recovery
- Make a risky movement decision

Mistakes should vary between matches.

Bots should not repeat identical action sequences.

9. Bot Personalities

Bots may have behavioral archetypes.

Examples:

Floater

- Prioritizes survival
- Avoids unnecessary risks
- Maintains controlled altitude
- Less aggressive item pursuit

Aggressor

- Takes more risks
- Pursues disruptive items
- Attempts to pressure the opponent
- May make more dramatic mistakes

Climber

- Prioritizes maximum height
- Uses upward momentum aggressively
- Can struggle when forced downward

Technician

- Consistent timing
- Strong recoveries
- Efficient item usage
- Lower mistake variance

Trickster

- Unpredictable movement
- Unusual item decisions
- Variable risk tolerance

Panicker

- Performs normally at first
- Makes more mistakes near the ground
- Has inconsistent recoveries

Bot personality should affect behavior, not raw cheating advantages.

10. Item Recognition and Reaction Time

Current bot item response can feel too immediate.

Bots need configurable perception, decision, and action delays.

The bot item interaction process should include:

1. Item enters awareness range.
2. Bot recognizes the item.
3. Bot evaluates the item.
4. Bot decides whether to pursue it.
5. Bot adjusts its movement.
6. Bot attempts pickup or activation.

Recommended Initial Reaction Ranges

Bot skill	Recognition and decision delay
Beginner	600–1,000 ms
Developing	450–800 ms
Intermediate	300–650 ms
Skilled	200–500 ms
Advanced	140–350 ms
Elite	100–280 ms

These values should be randomized.

No bot should respond at the exact same speed every time.

Item Awareness

Lower-skilled bots should:

- Have smaller awareness ranges
- Miss items outside their immediate movement path

- Misidentify item priority
- Ignore some items entirely

Higher-skilled bots should:

- Notice items from farther away
- Evaluate item value
- Consider survival risk
- Recognize opponent positioning
- Occasionally hesitate or make the wrong decision

Item Pursuit Probability

Bots should not pursue every useful item.

Example:

Beginner: Pursues 25–45% of recognized useful items

Intermediate: Pursues 45–70%

Skilled: Pursues 65–85%

Elite: Pursues 75–92%

The final decision should also consider:

- Distance
 - Current altitude
 - Current momentum
 - Item value
 - Opponent position
 - Risk of falling
 - Remaining item duration
 - Active abilities
 - Bot personality
-

11. Bots in Ranked Play

Bots may be used in ranked play, but restrictions are required.

Recommended policy:

Tier	Bot availability
Bronze	Allowed
Silver	Allowed
Gold	Allowed with reduced frequency
Platinum	Rare
Diamond	Human preferred, bots only as emergency backfill
Master	Human only
Grandmaster	Human only

Rating Rewards Against Bots

Recommended initial rules:

Bronze and Silver

- Normal rank-star progression
- Normal or slightly reduced Glicko gains
- Reduced rating loss for losing to a bot

Gold and Platinum

- Reduced Glicko gains
- Normal visible progression
- Bot frequency limited

Diamond

- Bots used only after extended queue times
- Reduced rank gain
- Reduced rank loss
- No leaderboard-critical advancement from repeatedly defeating bots

Master and Grandmaster

- Human opponents only
- No bot backfill

- Leaderboard placement based on human competitive matches

Anti-Farming

The system should prevent repeated farming of predictable bots.

Possible protections:

- Rotate bot personalities
 - Rotate bot loadouts
 - Vary performance parameters
 - Reduce rewards after repeated bot matches
 - Limit repeated matching against the same bot profile
 - Detect abnormal win patterns
 - Require human matches for upper-tier advancement
-

12. Placement Matches

At the beginning of a player's first ranked season, use placement matches.

Recommended count:

5 placement matches

Returning players may use fewer placement matches.

Placement matchmaking should use:

- Previous-season Glicko rating
- Historical survival performance
- Current rating deviation
- Placement match results

New players should begin with high rating uncertainty.

The system should adjust quickly if their demonstrated skill is much higher or lower than the starting estimate.

Placement matches may include bots at lower skill levels when no suitable player is available.

13. Seasonal Structure

Each competitive season lasts:

90 days

At the end of the season:

- Visible rank progression resets.
- Seasonal leaderboards close.
- Seasonal rewards are granted.
- A new season begins.
- Glicko rating is softly compressed rather than completely erased.
- Historical achievements remain permanently recorded.

Soft Rating Reset

Do not reset every player to the exact same rating.

Instead, pull ratings toward the center.

Conceptual example:

New rating = $1500 + ((\text{old rating} - 1500) \times 0.60)$

The compression value should be configurable.

This allows strong players to remain above average while giving everyone room to climb again.

Rating deviation should increase at the start of a new season so the system can quickly reassess skill.

Visible Rank Reset

Example seasonal placement:

Previous finish	New-season starting area
Bronze	Bronze
Silver	Bronze or Silver V

Gold	Silver
Platinum	Gold
Diamond	Platinum
Master	Diamond
Grandmaster	Diamond or Master placement

Exact placements should be determined through testing.

Players should not need to replay the entire beginner climb every season.

14. Seasonal Rewards

Rewards should be based on the highest rank reached during the season, not only the rank held on the final day.

Possible seasonal rewards:

- Coins
- Rank badges
- Profile borders
- Tap effects
- Character particles
- Victory animations
- Titles
- Seasonal character variants
- Seasonal backgrounds
- Leaderboard emblems

Rewards should remain cosmetic and should not create competitive advantages.

Example:

Rank reached	Reward category
Bronze	Coins and seasonal participation badge

Silver	Additional coins and profile badge
Gold	Seasonal tap effect
Platinum	Animated badge or profile frame
Diamond	Exclusive character particle effect
Master	Exclusive seasonal character variant
Grandmaster	Numbered leaderboard badge and premium cosmetic

The player should permanently retain a record of:

- Season number
- Highest rank
- Final rank
- Peak Glicko rating
- Win/loss record
- Best survival result
- Leaderboard placement
- Earned seasonal rewards

15. Win Streaks

Use win streak bonuses in lower and middle ranks.

Recommended behavior:

- Win streak begins after three consecutive wins.
- Bonus rank progression applies.
- Win streak bonus ends after a loss.
- Win streak bonuses stop at Diamond or Master.

Example:

1 normal win = 1 star

3+ win streak = 2 stars per win

Glicko calculations should still use the actual match results and should not be artificially doubled.

The bonus applies to visible rank progression only.

16. Match Result Processing

At the end of each ranked match:

1. Validate the match result.
2. Confirm that neither player disconnected improperly.
3. Record survival durations.
4. Record the winner and loser.
5. Update rolling performance statistics.
6. Calculate Glicko-2 updates.
7. Update visible rank stars or points.
8. Apply win streak changes.
9. Update seasonal statistics.
10. Update bot statistics if applicable.
11. Check promotion or demotion.
12. Grant milestone rewards.
13. Save match history.
14. Update leaderboards.

The rating update should occur server-side.

Clients must not be trusted to submit final rating values.

17. Disconnect and Abandonment Handling

Disconnect rules should distinguish technical issues from intentional quitting.

Recommended baseline:

- Reconnection grace period
- Match continues briefly during temporary disconnect
- Repeated abandonment counts as a loss
- Opponent receives the win when appropriate
- Abandonment rate is tracked

- Frequent intentional disconnects may result in queue penalties

Bots should not receive abnormal rating rewards because a player experienced a verified server-side failure.

The backend should log:

- Player disconnect timestamp
 - Reconnect attempt
 - Match state
 - Network error
 - Client closure
 - Server error
 - Final result reason
-

18. Leaderboards

Include seasonal leaderboards for:

- Highest Glicko rating
- Grandmaster ranking
- Highest average survival time
- Longest individual survival
- Most ranked wins
- Highest win streak
- Most improved rating
- Friends-only rankings

Only verified ranked matches should affect primary competitive leaderboards.

Bot matches should be excluded from certain prestige leaderboards or clearly categorized separately.

Master and Grandmaster leaderboard placement should prioritize human-versus-human results.

19. Required Data Model

Player Ranked Profile

player_id
season_id
glicko_rating
glicko_deviation
glicko_volatility
visible_tier
visible_division
rank_stars
highest_tier_season
highest_division_season
highest_tier_lifetime
highest_rating_season
highest_rating_lifetime
wins
losses
bot_wins
bot_losses
human_wins
human_losses
win_streak
best_win_streak
placement_matches_completed
placement_matches_required
ranked_status
updated_at

Player Performance Profile

player_id
average_survival_ms
median_survival_ms
recent_10_average_ms
recent_25_average_ms
best_survival_ms
recent_best_survival_ms
survival_variance
average_taps_per_second
recovery_success_rate
item_pickup_rate
item_use_success_rate
hazard_avoidance_rate
disconnect_rate
abandonment_rate

updated_at

Ranked Match Record

match_id
season_id
player_one_id
player_two_id
player_two_is_bot
bot_profile_id
player_one_rating_before
player_one_rating_after
player_two_rating_before
player_two_rating_after
player_one_survival_ms
player_two_survival_ms
winner_id
result_reason
queue_duration_ms
matchmaking_range
connection_quality
started_at
ended_at
validated

Bot Profile

bot_profile_id
display_name
personality
glicko_rating
target_survival_min_ms
target_survival_max_ms
reaction_delay_min_ms
reaction_delay_max_ms
item_awareness_radius
item_pursuit_probability
tap_accuracy
recovery_skill
error_frequency
risk_tolerance
active

updated_at

Season Record

season_id
season_name
season_number
start_date
end_date
status
rating_compression_factor
placement_match_count
reward_configuration
created_at

20. Administrative Configuration

The following values should be remotely configurable without requiring a client update:

- Season start and end dates
 - Rank tier thresholds
 - Stars required per division
 - Rank floors
 - Placement match count
 - Matchmaking search duration
 - Rating expansion intervals
 - Survival-time compatibility thresholds
 - Bot insertion timing
 - Bot reaction ranges
 - Bot item pursuit probability
 - Bot availability by rank
 - Ranked rewards against bots
 - Rating compression factor
 - Win streak requirements
 - Win streak bonus
 - Seasonal reward table
 - Disconnect grace period
 - Queue penalty duration
-

21. MVP Scope

The first implementation should include:

- Glicko-2 rating calculations
 - Bronze through Diamond tiers
 - Five divisions per tier
 - Master and Grandmaster placeholders
 - Rank stars
 - Rank floors
 - Five placement matches
 - Rolling average survival time
 - Recent 10-match survival average
 - Matchmaking using rating and survival compatibility
 - Expanding matchmaking search
 - Bot backfill
 - Three bot difficulty ranges
 - Randomized bot reaction delays
 - Slower and more human-like item responses
 - Basic seasonal reset
 - 90-day seasons
 - Seasonal rank rewards
 - Match history
 - Server-side result validation
 - Basic anti-farming protection
-

22. Post-MVP Scope

After the ranking foundation is stable, add:

- Full bot personalities
- Advanced performance profiling
- Friend leaderboards
- Regional leaderboards
- Grandmaster leaderboard
- Ghost opponents
- Replay support
- Match spectating
- Seasonal statistics pages
- Placement prediction

- Smurf detection
 - Advanced disconnect analysis
 - Dynamic bot calibration
 - Automated matchmaking tuning
 - Seasonal cosmetic reward tracks
 - Competitive tournaments
 - Rank decay for inactive Master and Grandmaster players
-

23. Acceptance Criteria

The system is ready for initial testing when:

- Two real players can complete a ranked match.
 - Both ratings update correctly through Glicko-2.
 - The winner generally gains rating.
 - The loser generally loses rating.
 - Rating uncertainty affects the size of rating changes.
 - Players receive visible rank progression.
 - Rank floors function correctly.
 - A 15-second player is not normally matched against a 72-second player.
 - A 63-second player may reasonably match against a 72-second player.
 - Matchmaking expands gradually.
 - A bot is inserted when no suitable player is found.
 - Bot difficulty corresponds with the player's rating.
 - Bots do not instantly react to every item.
 - Bot item recognition includes randomized delays.
 - Bots occasionally miss, ignore, or mishandle items.
 - Bot matches are visibly identified.
 - Bot farming protections function.
 - Ranked progress resets after 90 days.
 - Historical seasonal accomplishments remain visible.
 - Seasonal rewards are granted based on the highest rank reached.
 - Rating updates and match validation occur server-side.
-

Final Design Principle

The competitive system should not attempt to give every player an easy match.

It should attempt to give every player a believable, fair, and motivating match.

The ideal opponent is not someone the player can defeat automatically.

The ideal opponent is someone who makes the player think:

I almost had them. One stronger run and I can win.